

ECG-NET: A deep LSTM autoencoder for detecting anomalous ECG

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ARTICLE INFO

Keywords:

ECG arrhythmia

Deep learning

LSTM

Autoencoder

Thresholding

Kapur's thresholding

ABSTRACT

The electrocardiogram (ECG) is a standard test to monitor the activity of the heart. Many cardiac abnormalities are manifested in the ECG including arrhythmia that refers to an abnormal heart rhythm. The basis of arrhythmia diagnosis is the identification of normal versus abnormal heart beats, and their correct classification based on ECG morphology. This paper proposes a novel and robust approach for representation learning of ECG sequences using a LSTM autoencoder for anomaly detection.

The encoder part encodes the ECG signal into a lower dimensional latent space representation and decoder part then tries to reconstruct the specified ECG signal. The model is trained only on normal (non-anomalous) ECG signals. Then reconstruction loss of test ECG signals are calculated. Next a reconstruction loss threshold value is determined from the frequency distribution of the reconstruction losses so that from the reconstruction loss value above a certain threshold is determined as anomaly, otherwise it will be treated as normal. Determination of threshold is done using manual and Kapur's automated thresholding procedures.

The aforementioned model has been applied on publicly available ECG5000 dataset. From the experimental results it is observed that the proposed model achieved more than 98% accuracy having precision, recall and F1 values more than 0.94, 0.97, 0.96 respectively. The performance of the proposed method is also found to be superior in most of the cases as compared to the results of seven other recent counter-part methods reported in the literature.

1. Introduction

Electrocardiogram (ECG or EKG) signal is a time series data sequence which generates a graphical representation of electrical impulses from myocardium (Fleming, 2012). Cardiac activity of the heart beat is recorded in the form of electrical signals in the ECG. It is recorded from many electrodes placed over chest skin during testing. The majority of the physicians prefer to use ECG signal as a primary screening tool to detect and diagnose cardiac diseases

ECG is composed of five different morphological waveforms, i.e., P, Q, R, S and T wave (Sivaraks and Ratanamahatana, 2015). The PQRSST waveform is produced through depolarization (working phase) and repolarization (resting phase) activity occurring in each cardiac cycle. The waveform is depicted in Fig. 1. The atrial depolarization wave is known as the P wave and is caused by the sinoatrial node (SA) which is considered to be the natural pacemaker of the heart responsible to generate the pulses. Next one is QRS complex that is combination of Q wave, R wave and S wave. Atrioventricular node (AV) generates this depolarization waveforms. This complex represents the depolarization of the ventricles (Sivaraks and Ratanamahatana, 2015). Last

wave of the cardiac cycle i.e., T wave indicates ventricular repolarization (Sivaraks and Ratanamahatana, 2015). A single cardiac cycle is composed of depolarization and repolarization from an atrium to a ventricle (Sivaraks and Ratanamahatana, 2015). ECG reports from various leads (generally, 12 leads) may be morphologically different depending on the vector of the heart.

A normal ECG report contains a consistent heart rhythm whereas an abnormal ECG does not follow any consistent rhythmic pattern. The anomaly ECG represents damage of the heart muscle or heart's rhythmical problems. The ECG wave and one interval of a single Heart Cycle in a normal ECG is shown in Fig. 1.

Traditionally, medical practitioners check whether the ECG report is normal or has some abnormalities by observing the ECG signal's rhythmic patterns by their expertise and experience. However, privacy regulation of personal healthcare records, the need of trained expert annotators, cost of collecting as well as annotating ECGs and variability of ECGs make the dataset creation a challenging task. Moreover, trained healthcare professionals/doctors may not be available always in emergency situations particularly in remote locations. In this context,

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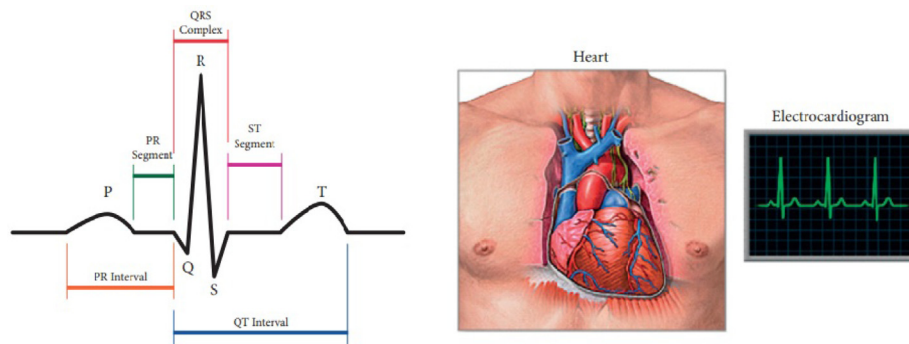


Fig. 1. The waves and intervals of a single Heart Cycle in a normal electrocardiogram (ECG).

automated prediction of arrhythmia from ECG signals is an important task of engineering applications of artificial intelligence to combat cardiac disease. Additionally, for developing a robust and scalable machine learning/deep learning based model for anomaly detection in ECGs, a large annotated ECG dataset is needed. Therefore, design and development of robust algorithms for detecting arrhythmia from ECGs automatically, is a need of the time.

Abnormality predictions from ECG signals using ML and DL models have drawn significant attention of the researchers in recent time. In recent years, various machine learning techniques with soft computing methods like fuzzy logic (Farhan et al., 2018), genetic algorithm (Diker et al., 2019), clustering (Zhang and Yue, 2017), etc. have been used in ECG classification and anomaly detection.

DL is a branch of ML and most of the DL methods provide a lot of promises compared to traditional ML such as better accuracy, significant flexibility, stronger generalization and less dependency as DL methods learn the inherent/features automatically from the data without the need of feature engineering to extract and select appropriate features for the given task. So, comparing with traditional machine learning (ML) methods, deep learning (DL) methods show stronger learning ability and can achieve higher accuracy. Handling real-world time series data is the biggest challenge (Gamboa, 2017) as most of the real-world time series datasets contain incomplete or missing values, as well as it suffers from noise. DL methods have been successfully applied in various time series data analysis tasks including electrocardiogram (ECG) (Murat et al., 2020), electroencephalogram (EEG) (Craik et al., 2019), stock market fluctuations (Khan et al., 2022), weekly weather reports prediction (Ren et al., 2021) and other time-sequence dependent phenomena.

Researcher studied that ECG time series analysis using DL methods perform well in general (Rajkumar et al., 2019). For automatic detection of ECG arrhythmia from ECG signals, several Deep Learning techniques like Convolutional Neural Network (CNN) (Acharya et al., 2017), Recurrent Neural Network (Singh et al., 2018), Long Short-Term Memory (LSTM) (Ji et al., 2021), Bidirectional Long Short-Term Memory (BiLSTM) (Lynn et al., 2021), Bidirectional Gated Recurrent Unit (BiGRU) (Lynn et al., 2019) and autoencoder (Matias et al., 2021) have been proposed.

In this paper, a novel LSTM based autoencoder architecture is proposed where an autoencoder is designed using three layer based LSTM model to detect the arrhythmia or anomaly ECG signals.

Autoencoder is the process that is trained to copy its input to its output. In the first step, it encodes the input image, pattern or signal and convert the input signal into a lower dimensional latent representation. In the next step, the latent representation is decoded to get back to that original image, pattern, signal as best as possible. Some notable applications of autoencoder are pattern recognition (Noering et al., 2021), anomaly detection (Zhang et al., 2019), feature extraction (Maggipinto et al., 2018), dimensionality reduction (Wang et al., 2016), image denoising (Bajaj et al., 2020) etc.

The proposed autoencoder architecture in this article, comprises of three LSTMs. LSTM networks are designed specifically to overcome the long-term dependency problem faced by RNNs (due to the vanishing/exploding gradient problem) (Hochreiter and Schmidhuber, 1997; Hochreiter, 1998). The challenge of accurately and reliably storing the knowledge of inputs from the remote past in Recurrent Neural Networks (RNNs) is known as the “long-term dependency problem” (Hochreiter and Schmidhuber, 1997; Hochreiter, 1998). In a sequence data analysis task, we know that the context of the previous inputs are important in order to predict future events. Learning the dependencies between past and present inputs by RNNs becomes difficult when the context is far away in the past. This happens because of the vanishing or exploding gradient problem, during the back propagation process in the training phase, where the gradients may become very small/even disappear or may become infinitely large (Hochreiter and Schmidhuber, 1997). To store the information about past events, memory cells have been introduced in LSTM, which can store the information over a longer period of time (Hochreiter and Schmidhuber, 1997). Depending on the requirement, this information needs to be forgotten or remembered selectively by the network. This is done by introducing various gates (viz., forget gate, input gate, output gate) in the LSTMs which can regulate the flow of information into and out of the memory cell as needed. This allows LSTM not only to preserve the important information as long as needed, but also prevents vanishing gradients as well as exploding gradients. LSTMs have feedback connections which make them different to more traditional feed forward neural networks. This property enables LSTMs to process entire sequences of data (e.g., time series) without treating each point in the sequence independently, but rather, retaining useful information about previous data in the sequence to help with the processing of new data points. As a result, LSTMs are particularly good at processing sequences of data such as text, speech and general time-series (Malhotra et al., 2015).

Challenges involved in automated ECG arrhythmia detection include but not limited to the following: (i) limited availability of the annotated ECG signals to train the model, (ii) involvement of data imbalance problem in ECG dataset, where the normal ECG signals are predominant over anomaly ECGs. The above said challenges are judiciously handled in the proposed (LSTM based autoencoder) ECG-NET method by training the model with only normal ECG signals (where no anomalous ECG signals are required during the training phase) and even then can achieve good accuracy during the testing phase considering both normal and anomalous test ECG signals. Thus, data imbalance problems and limited availability of annotated samples (particularly anomalous ECG signals) are handled.

The major contribution of the proposed method may be summarized as follows: (i) the proposed ECG-NET requires only normal ECG signal to train the model without the need of anomalous signals which may be very beneficial when the anomalous signals are not available a priori to build the model, (ii) data imbalance problems and limited availability of annotated samples (particularly anomalous ECG signals) are handled

by the proposed method, (iii) proposed an automated reconstruction loss threshold selection approach on testing phase based on Kapur's histogram thresholding approach, (iv) also proposed three LSTM based autoencoder architecture which yields better accuracy than existing architectures.

The rest of the article is organized as follows. Section 2 describes the related works for detecting the anomaly ECG signals. In Section 3, background studies are highlighted. The proposed method and the basic architecture of our proposed method are elaborated in Section 4. In Section 5, experimental evaluation describes the used dataset, data exploration, tensor creation, operating environment and hyperparameters settings. Section 6 highlights the experimental results and analysis. Finally, conclusions and future work are drawn in Section 7.

2. Related work

Automatic anomaly detection from various time series data like electrocardiogram (ECG), electroencephalogram (EEG) using DL methods are gaining popularity in recent years. Various unsupervised (Pereira and Silveira, 2019), supervised (Acar et al., 1997) and semi-supervised (Zhai et al., 2020) deep learning models for detecting abnormalities in heart beat have been proposed by different researchers.

In Chauhan and Vig (2015), proposed deep LSTM neural networks to detect anomaly in ECG sequences data. The advantage of this approach is that it does not require prior knowledge of abnormalities to work. Another LSTM based model LSTM-AD has been proposed by Malhotra et al. (2015) to detect the anomaly in time series data. The presented approach (Malhotra et al., 2015) uses a predictor that learns the behavior of normal heart beat pattern, which are used for detecting the abnormal heart beat pattern. Another DL model called TimeNet, has been proposed by Malhotra et al. (2017) (Gupta et al., 2018) which uses a sequence autoencoder for extracting time sequence features from ECG data.

In Dutta et al. (2021), Dutta et al. proposed MED-NET to detect ECG anomaly detection. The proposed LSTM based autoencoder model works on the concept of compression and reconstruction of ECG signals. Encoder is used to compress normal ECG signals. The reconstruction of the ECG signals is done by the decoder part of the proposed architecture. The model is trained with only normal ECG samples and unknown ECG signal is predicted either as normal or not normal (i.e. abnormal). The predication of the normal ECG as well as anomaly ECG is done on the basis of a threshold collected from the training phase. This MED-NET model gains the overall accuracy of 97.93%.

In Noor et al. (2021), Noor et al. proposed RNN-LSTM autoencoder to predict the normal ECG and risky ECG. For training purpose, at first the signals are passed through one RNN output of which is then fed to LSTM autoencoder to predict normal and risky (Premature Ventricular Contraction(PVC) only). It is also a threshold based prediction and the method is trained with only normal ECG sample. Based on the training, threshold is selected. The accuracy of RNN-LSTM autoencoder is 97.24%. Although it predicts the risky and normal ECG heartbeat is quite significantly but the testing set is composed of only 8.76% (438 out of 5000) ECG samples of the total dataset.

Pereira and Silveira (2019) proposed three unsupervised model to detect ECG anomaly. All three proposed models use variational autoencoder in the first phase, whereas the second part of the model uses SVM, variational autoencoder with cluster, variational autoencoder with wasserstein distance similarity and the accuracy of those models are 98.43%, 95.96% and 95.96% respectively.

2.1. Difference between proposed method and other counterpart methods

Although the LSTM based autoencoder is also used in Dutta et al. (2021), but our proposed model is structurally different as it uses three LSTM layers in each encoder and decoder block to better capture the encoded feature representation as compared to Dutta et al. (2021)

where two LSTM layers are used in encoder and decoder part. Moreover (unlike Dutta et al. method Dutta et al., 2021), in the proposed method dropout operations are also applied in order to control the regularization factor by not allowing the model to be gone to underfitting or overfitting situation particularly for unseen (test) cases. Additionally, in the proposed method two (one automatic and another manual) different reconstruction loss threshold selection procedures are adapted in the testing phase on both normal and anomaly ECG signals, whereas authors in Dutta et al. (2021) selected the reconstruction loss threshold on the training phase for normal ECG signal only.

Various other approaches like combining other model with LSTM autoencoder have also been proposed for predicting anomaly ECG. One such approach is proposed by Noor et al. (2021), where RNN is used along with LSTM autoencoder. In comparison with Noor et al. (2021), our proposed model does not use any additional RNN architecture, which reduces the resource overhead. Furthermore (unlike Noor et al.), the dropout method is applied in the first two layers of both encoder block as well as decoder block in the proposed method to reduce the overfitting or underfitting problems. For extracting the non-linear features of ECG signals, our proposed method uses ReLU as the activation function in first two layers of both encoder block as well as decoder block in autoencoder architecture. No such information about using any activation function is present in Noor et al. (2021) literature. Though, both Noor et al. and the proposed method uses the manual threshold procedure based on the reconstruction loss of ECG signals, but they differs from the origin of selecting threshold value. Our proposed method selects the threshold value based on reconstruction loss of testing ECG, whereas Noor et al. used trained ECG samples for yielding threshold value. Additionally, our proposed method is also performed automated thresholding procedure for predicting normal ECG as well as anomaly ECG signals.

It is worth noting that, unlike Murat et al. work (Murat et al., 2020), the proposed autoencoder based ECG-NET method does not require to have anomalous signals during the training phase of the autoencoder; the training is done using only normal ECG signals which is a significant advantage over the Murat method (that requires to have sufficient number of training samples from all the classes including abnormal ones). In real life scenarios, the annotated anomalous signals may not always be available during the training process while building an ECG prediction model, however, it requires to predict normal and anomalous signals both. In that situation, Murat et al. method (Murat et al., 2020) will not work at all, however, the proposed ECG-NET method will work.

Researchers also focused on fusing traditional machine learning with deep learning approach. Pereira and Silveira (2019) shows how three different traditional machine learning methods viz., SVM, wasserstein distance similarity and clustering can be used with a deep learning model i.e., variational autoencoder for detecting ECG abnormalities.

3. Background study

In this ECG-NET proposed model, LSTM is utilized inside the autoencoder. In this section, autoencoder and LSTM is defined briefly.

3.1. Autoencoder

Autoencoder(AE) is a neural network architecture which is capable to discover the structure of dataset in order to develop a compressed representation of the input data. Various types of autoencoder architecture exist for learning and ensuring the compressed representation represents the meaningful features of the original data input. This form of learning is not intended to produce a model capable of making accurate predictions on new data, but is used to study the characteristics of data and to extract features or patterns. The neural networks of autoencoder are used to learn the representation of input data. The predominant goal of the autoencoder is to reconstruct the output as

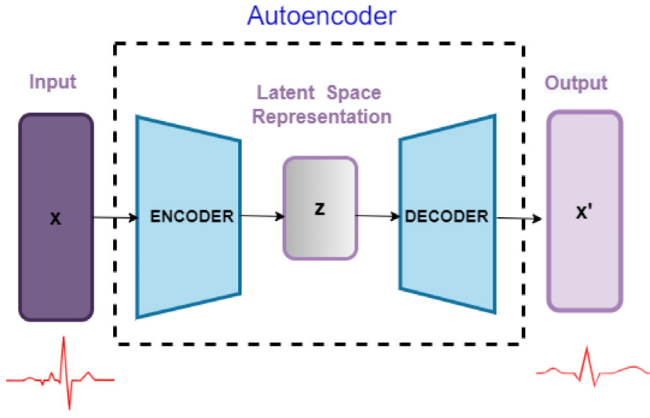


Fig. 2. Schema of a basic autoencoder.

close as possible to the input data. It composed with two parts viz., encoder and decoder.

The first part (i.e., encoder) performs a series of data transformations on the input data $x \in R^m$ to lower dimensional (i.e., compressed domain) representation called bottleneck layer or *latent space representation* $z \in R^n$ (where $n < m$) that consists of the main and important features of the original input. So, the data compression procedure is performed in encoder part.

Whereas in the second part (i.e., decoder) performs the inverse transformations of the encoder onto the original input data to produce the decoded output $x' \in R^q$ (where $q > n$). Decoder part reconstructs the input signal from the *latent space representation* (z) (produced by the encoder part) into preferred dimension as close as possible to the original input data.

Here, m , n , q represent the dimensions of the input signal x , latent space representation z and decoded output x' respectively. Please note that, for the present problem dimension q of the decoded output is same as the dimension m of the input data.

The basic architecture of an autoencoder is shown in Fig. 2 (Hinton and Salakhutdinov, 2006).

3.2. Long short term memory

Typical recurrent neural networks (RNN) are capable of handling sequential data by accepting the current input data as well as memorizing the previously received inputs. Though RNNs show significant capabilities to handle time series data, but suffers from vanishing and exploding gradient problem. Hochreiter (1998) showed that this problem can be solved via the use of multiplicative input, output and forget gates. These various gates are used for preserving state information. This Long short-term memory (LSTM) network architecture is capable of preventing the output of a neural network from either decaying (long-term dependency) or exploding as it passes through the feedback loops for the given input. Such feedback loops allow the network to perform better at recognizing sequential patterns compared to other deep convolution neural networks. Due to the ability to learn long-term dependency, LSTMs are applicable to numerous long sequence learning problems such as language modeling, machine translation, and many other related tasks.

A single typical LSTM unit is made up of a cell state and three gates, viz., forget gate, input gate, and output gate. In addition to these, an LSTM also has hidden states (working memory) associated with gates which are used for carrying information from immediately previous events and to overwrite at every step. The overall architecture of a basic LSTM unit is represented in Fig. 3 (Hochreiter and Schmidhuber, 1997). Here, h_{t-1} represents the hidden state of the previous timestamp whereas h_t represents the hidden state of the current timestamp. C_{t-1}

represents cell state from previous timestamp and cell state of the current time stamp is denoted by C_t . Fig. 3 depicts that different activation functions have been applied to different gates to incorporate different nonlinear transformations. Brief introduction of different gates used in LSTM are described below.

• Forget Gate

As the name suggests, this gate decides whether an information it should remember or to forget. It determines what is important to retain from the previous cell state. The decision of retaining or forgetting the information is broadly dependent on the output of the activation function. If the output value is close to 0, it means to forget, and if it is close to 1, it means to keep that information. Forget gate takes previous hidden state h_{t-1} and the new input data x_t as inputs. The sigmoid activation function is applied to these two inputs for getting the output f_t value within [0, 1] interval. Here, the cell performs point-by-point multiplication. The overall function and equation of forget gate is shown in Fig. 3. Mathematically the output operation (Hochreiter and Schmidhuber, 1997) of the forget gate can be expressed as:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

where, x_t is the input to the current timestamp t ; W_f represents the weight matrix associated with input and hidden state; h_{t-1} depicts the hidden state of the previous timestamp and b_f is the bias with respect to W_f .

• Input Gate

Input gate helps in deciding the importance of the information by updating the cell state. It is a decider state that helps to decide what new information it needs to store in the cell state. First, a sigmoid layer known as the input gate layer i_t is used to decide what value it should update by mapping it in the interval [0, 1]. Next, a tanh layer creates a vector of recent candidate values C_t . Sigmoid decides the weight of information while tanh reduces the bias of the network. The output values generated from the activation functions viz., sigmoid and tanh functions are ready for point-by-point multiplication. The representation and equation of input gate is demonstrated in Fig. 3 and the mathematical operations (Hochreiter and Schmidhuber, 1997) of input gate is calculated as:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (3)$$

where,

X_t : Input at the current timestamp t , h_{t-1} : Hidden state from the previous timestamp t , W_i : Weight parameter of sigmoid operator associated with input and hidden state, b_i : Corresponding bias with respect to W_i , C_t : Value generated by tanh, W_c : Weight parameter of tanh operator associated with input and hidden state and b_c : Corresponding bias with respect to W_c .

• Cell State

For governing the input, as well as to forget the irrelevant information, LSTM uses two dedicated gates. The input gate i_t governs how much will it take new information under consideration via $\tilde{C}_t$ while the forget gate forwards the information about the retention of old content coming via (C_{t-1}) . This is done by point-wise multiplication technique. After that, the multiplied terms perform point-by-point addition. Updation of C_t is done by following update equation and the overall scenario is depicted in Fig. 3. The equation of cell state (Hochreiter and Schmidhuber, 1997) can be expressed as:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (4)$$

where, f_t : content of forget gate at timestamp t , i_t : content of input gate at timestamp t , C_t : Value generated by tanh and C_{t-1} :

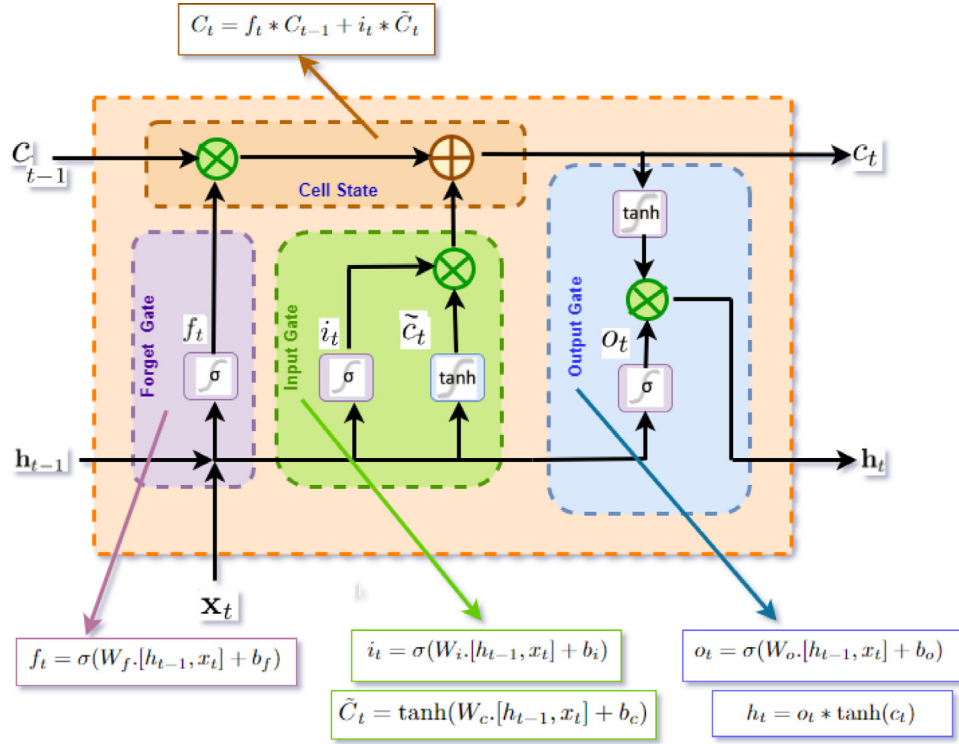


Fig. 3. Basic LSTM Unit and corresponding individual blocks.

Content from the previous timestamp $t - 1$. If the forget gate gives value approximately 1 and the input gate gives approximate value 0, then the past memory cells will be saved over time and will pass to the current time step. This idea helps to tackle the vanishing gradient problem and also captures long range dependencies within sequence.

• Output Gate

The final or the output gate helps in deciding the values of next hidden state of the network. Information goes through the sigmoid function determines whether to allow 0 or 1 values. Updated cell content from the cell state goes to the tanh function which assigns weights to the values provided. This is done by determining their relevance on a scale of $[-1, 1]$ and then it gets point-by-point multiplied with the sigmoid output. This final stage of the architecture helps the hidden state to decide which information it should carry. The representation and equation of final output gate is shown in Fig. 3. Equation of the output gate (Hochreiter and Schmidhuber, 1997) is as follows:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (5)$$

$$h_t = o_t * \tanh(c_t) \quad (6)$$

where, o_t : Content of output gate at timestamp t , h_t : Final output, W_o : Weight parameter of Output gate and b_o : Corresponding bias with respect to W_o .

4. Proposed method

The block diagram of the proposed LSTM based autoencoder model “ECG-NET” for ECG anomaly detection is depicted in Fig. 4. The detailed description of the overall process are described in the following subsections.

Firstly, the ECG signal is fed to the model as input. Then data preprocessing is done on the input signals in order to segment the dataset into small time intervals (note that, in our case we used already segmented ECG dataset having 140 intervals) and the data is further

split into training, validation and testing sets to feed subsequently into the proposed LSTM based autoencoder model. The core architecture of the proposed method relies on the LSTM based autoencoder approach having 3 LSTM layers in each encoder and decoder block as shown in Fig. 5. The encoder part compresses the ECG signal into a lower dimensional latent space representation which is being used by the decoder part to reconstruct the original ECG signal. Here, we pass only the normal ECG training dataset through LSTM autoencoder block in order to generate the encoded latent space representation of the normal ECG signals. The idea of using only the normal ECG signal in the training phase is that the reconstruction done by the decoder part from the latent space representation of the normal ECG signal will have less reconstruction loss as compared to the anomaly signals (in the testing case). Thus, once the training phase is over, the test samples (both normal and anomaly ECG signals) are passed to generate the reconstructed ECG samples and then reconstruction loss values are calculated. The frequency distribution of reconstruction loss of each trained ECG samples are plotted as histogram which is shown in Fig. 6. From the frequency distribution curve of the reconstruction loss values, an optimum threshold value τ is determined. We employed one manual (visual observation) based threshold determination technique from the histogram of reconstruction loss and another automated thresholding technique borrowing the idea from Kapur’s histogram thresholding technique (Kapur et al., 1985). Determined τ value is thus used for classification of any test ECG signal. If the reconstruction loss falls within the threshold τ , then that ECG signal is predicted as normal, otherwise it is detected as anomaly ECG as depicted in Fig. 6.

The detailed procedure of each block is described below.

4.1. Data preprocessing

Signal preprocessing is an important step in biomedical research due to the presence of noise, artifacts and any trends that can influence the results. The following subsections describe the data preprocessing of ECG signals.

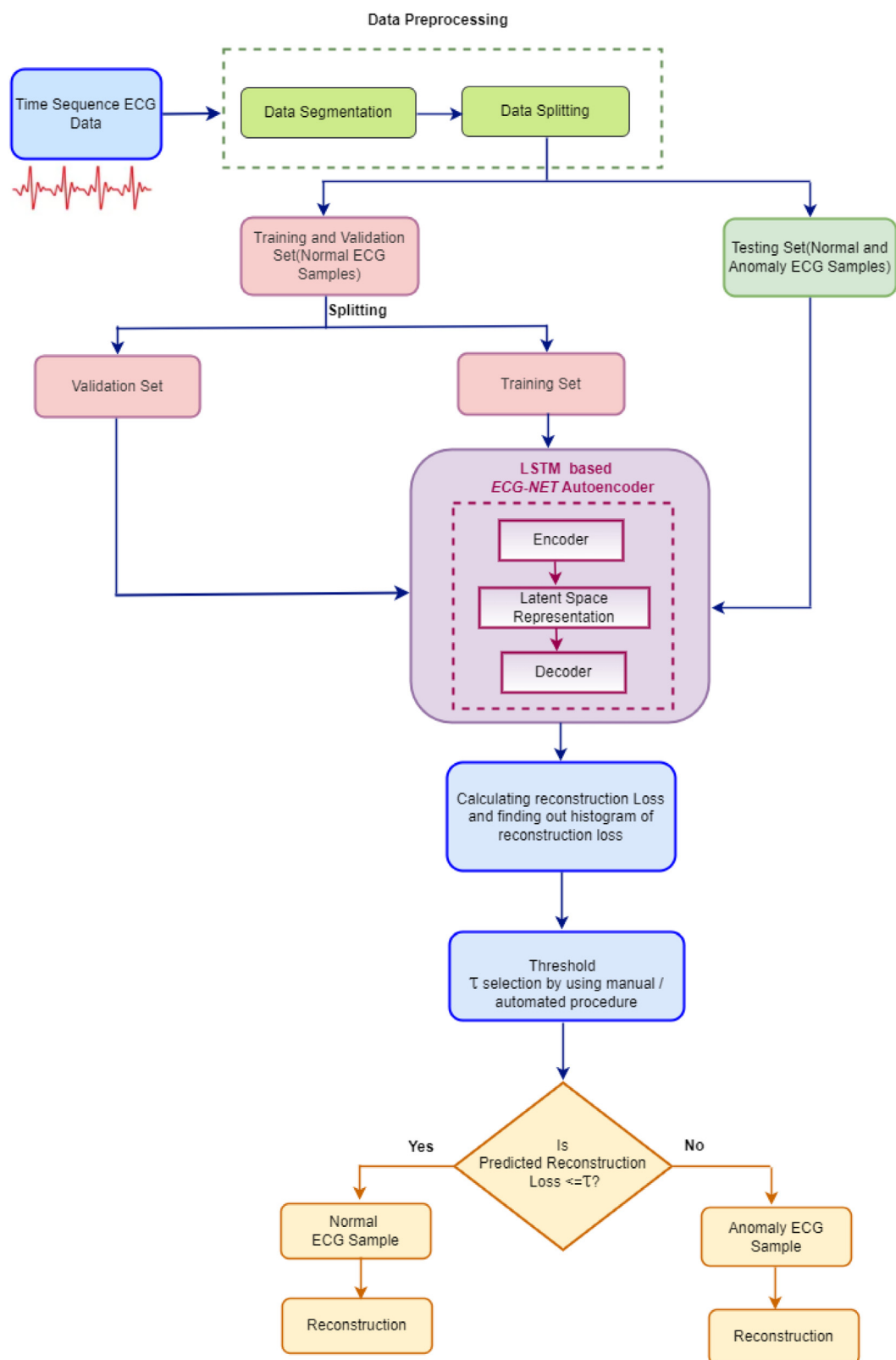


Fig. 4. Block diagram of ECG-NET architecture.

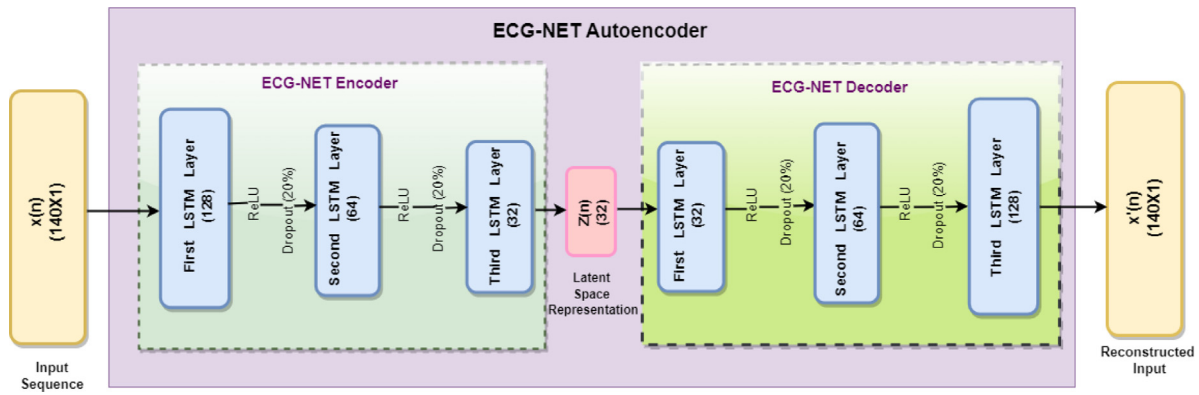


Fig. 5. Architecture of the proposed ECG-NET model.

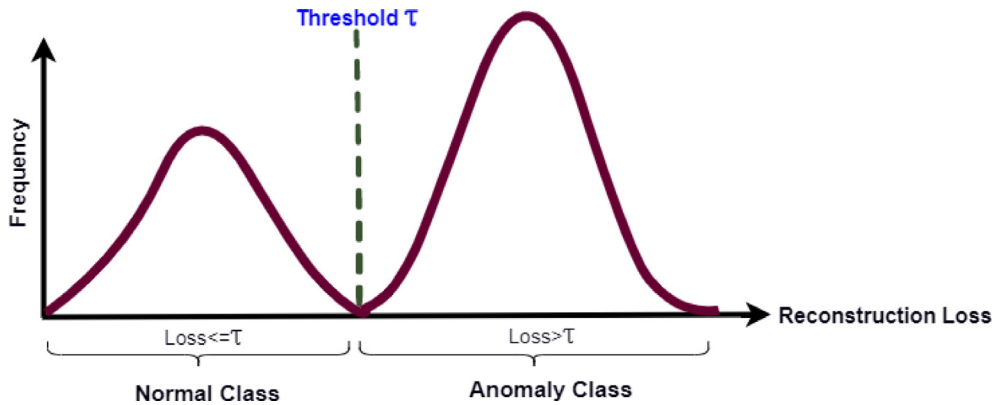


Fig. 6. Reconstruction loss threshold selection policy based normal and anomaly classes reconstruction loss histogram.

4.1.1. Data segmentation

ECG signals are captured continuously from the 12 different leads those are attached on different places of the patient's skin. ECG signal is quasi-periodic voltage signal which represents the electrical activity of the heart. Many researchers have proposed various segmentation methods (Gurve et al., 2016) of the ECG signals with respect to time duration. Each segment of ECG i.e., P wave, QRS complex and T wave are extracted into discrete numeric values. After segmentation, the graphical representation of the ECG signal is converted to two dimensional (2D) dataset. The row dimension of the two dimensional matrix represents the number of ECG samples and the column dimension shows the sequence of features in a particular ECG report. Please note that, in this work the ECG dataset which we considered is already segmented into 140 time steps and publicly available in "UCR Time Series Classification".¹

4.1.2. Data splitting

As mentioned earlier, in the training phase of the proposed LSTM based autoencoder method, only normal ECG signals are taken into consideration to learn the latent space feature representation of the normal ECG signals. As the model is trained using only normal ECG samples, therefore both the training set and validation set contain only the normal ECG samples. However, needless to mention here that both normal and anomaly ECG samples are tested in testing phase of the proposed method (as described earlier) to observe efficacy of the proposed architecture. The number of normal ECG signals used in training and validation phase are 2101 and 234 respectively. Whereas, 584 and 2081 number of normal and anomaly ECG signals are respectively used in the test case.

4.2. Proposed LSTM autoencoder architecture

The proposed model uses three-layered LSTM autoencoder architecture. The LSTM based autoencoder has two major blocks. The first block is the LSTM-Encoder block which is followed by LSTM-Decoder block in the second. It is empirically seen that the proposed architecture is able to produce optimal accuracy when three LSTM layers are used. We have experimented with different LSTM models with varying layers and found that three layers architecture performs the best for the present problem.

The detailed architecture of the LSTM based autoencoder model ECG-NET is shown in Fig. 5 and described in the following subsections.

4.2.1. LSTM-Encoder

The proposed encoder module consists of three LSTM layers. Each ECG data recorded the heart beat pattern of 140 continuous time step. This normal ECG time series data (140×1) is fed into the first LSTM layer. To capture the temporal dependency of the data. The first LSTM layer uses 128 hidden neurons. The sequences returned by these hidden states are then undergone through the Rectified Linear Unit (ReLU) activation function. These encoded representations hold the non-linear temporal features which have been learned so far from the ECG signal. After applying activation function ReLU, dropout(20%) is applied. The sequential data is further encoded with the help of second LSTM layer which now consists of 64 hidden neurons. For further extraction of non linear features from the time series based ECG data, another activation function ReLU is used followed by a 20% dropout operations. In the last step of the encoder module, these sequential data is further encoded by third LSTM layer having 32 hidden neurons. This module produces the final encoded latent space representation of the original ECG signal.

¹ <http://www.timeseriesclassification.com/description.php?Dataset=ECG5000>.

4.2.2. LSTM-Decoder

As similar to LSTM-Encoder block, the LSTM-Decoder part also has three LSTM layers. Actually, it is the mirror image of the encoder module and thus the decoder layers stacked in the reverse order as of the encoder block. The aim of this decoder module is to reconstruct the original data from the encoded latent space representation signals which is available after the LSTM based encoder module. As the decoder module is also a feed forward network, the first LSTM of the decoder module takes the encoded representation from the final layer of the encoder module. The first LSTM layer consists of 32 hidden neurons. Here also the activation function ReLU (Rectified Linear Unit) is used for extracting the non-linear temporal features from the signal and dropout (20%) is used. In the second LSTM layer, the output of the first layer is fed into 64 hidden LSTM layer. Like previously, again the activation function ReLU and dropout (20%) are used. Before the reconstruction of the ECG signal another LSTM layer with 128 hidden neurons is used. At the end of the decoding operation, the model comes up with the reconstructed ECG signal from encoded form.

4.3. Threshold selection

After training the model, the threshold value τ is chosen and prediction is performed based on the threshold value τ . The following subsections illustrate the procedure of threshold τ selection of tested ECG signals.

For detecting abnormality in ECG, the proposed approach *ECG-NET* relies on the concept of selecting threshold value τ . Based on the training, the optimum threshold value τ is selected. The normal ECG signals are trained with a suitable number of epochs. We are interested to observe the cumulative distribution of loss obtained from the training phase. The selection of threshold value τ can be done manually or by automated threshold selection procedure. In the proposed work, we have selected threshold by both manual and automated procedure. The following subsections will briefly describe the procedures of both threshold selection.

4.3.1. Manual thresholding

As stated earlier, the histogram (i.e., number of occurrence or frequency of each reconstruction loss) of the test data achieved by the proposed method are plotted to manually determine the appropriate threshold value τ , so that the reconstructed ECG signals with loss values below the threshold value τ are classified as normal and above which are classified as anomaly ECG. From the histogram of reconstruction loss, we fixed the threshold value τ , such that, it finds the valley portion of two peaks as depicted in Fig. 6.

4.3.2. Automated thresholding

Here, in this proposed architecture, we have used Kapur's threshold method (Kapur et al., 1985) as automated thresholding technique applied on probability distribution curve of the reconstruction losses achieved from the test data sets (containing both normal and anomaly ECG signals).

Kapur's thresholding method is an automated thresholding method that is used in image processing. This is an entropy-based efficient thresholding technique for image segmentation. In this thresholding method, here two probability distributions ω_n and ω_a are considered respectively for normal and anomaly ECG reconstruction loss histograms of the test ECG samples as shown in Fig. 6. Let $p_0, p_1, \dots, p_{(L-1)}$ be the probability distribution of loss values associated with both normal and anomaly ECG reconstruction, assuming that the reconstruction loss values are in the range of 0 to $(L-1)$. Here, $(L-1)$ presents the highest reconstruction loss of ECG signals. Now from this entire distribution, we derive two probability distributions, one for discrete values 0 to t and the other for values $(t+1)$ to $(L-1)$. p_{ω_n} and p_{ω_a} denote the total frequency of reconstruction loss values falls with in the range of 0 to t and $(t+1)$ to $(L-1)$ respectively. Assuming a reconstruction loss

threshold value t , two distributions ω_n and ω_a can be defined as follows:

$$\omega_n : \frac{p_0}{p_{\omega_n}}, \frac{p_1}{p_{\omega_n}}, \dots, \frac{p_t}{p_{\omega_n}}, \text{ and } \omega_a : \frac{p_{t+1}}{p_{\omega_a}}, \frac{p_{t+2}}{p_{\omega_a}}, \dots, \frac{p_{L-1}}{p_{\omega_a}}. \quad (7)$$

The entropies of normal and anomaly classes (for reconstruction loss threshold value t) are calculated in Eq. (8).

$$H_{\omega_n}(t) = - \sum_{i=0}^t \frac{p_i}{p_{\omega_n}} \log_2 \frac{p_i}{p_{\omega_n}}, \text{ and } H_{\omega_a}(t) = - \sum_{i=t+1}^{L-1} \frac{p_i}{p_{\omega_a}} \log_2 \frac{p_i}{p_{\omega_a}} \quad (8)$$

The optimal reconstruction loss threshold value τ is determined by selecting the threshold t for which summation of these two entropy values (i.e., $H_{\omega_n}(t) + H_{\omega_a}(t)$) becomes maximum as shown in Eq. (9).

$$\tau = \arg \max_t [H_{\omega_n}(t) + H_{\omega_a}(t)] \quad (9)$$

One of the novelties in the proposed model is to choose the reconstruction loss value appropriately either by manual thresholding method or by automated thresholding method using Kapur's technique. When the histogram of the reconstruction loss is having a clear-cut valley portion between two peaks, we can go for manual thresholding procedure as shown in Fig. 6. However, when there is no clear cut valley available in the histogram of the reconstruction loss, the manual determination of the threshold would be difficult. In that case, we can go for an automated thresholding procedure such as Kapur adopted in this paper. In general, the automated threshold selection procedure using Kapur's method can be useful immaterial of the fact that whether the valley is present or not in the reconstruction loss histogram. Therefore, a general solution for thresholding the reconstruction loss histogram problem can be addressed by adopting automated thresholding using Kapur's method as done in this article.

4.4. Prediction of normal and anomaly ECG

Based on the selected threshold τ (obtained as described in Section 4.3), the model predicts normal and abnormal ECG test samples as follows. The reconstruction loss of the test ECG sample falls under (i.e., less than or equal to the selected threshold value τ) is considered as normal ECG samples. On the other hand, the reconstruction loss of the test ECG sample having value more than τ is considered as anomaly ECG signal. This is because of the fact that when an abnormal test ECG sample is supplied to the trained model, it tries to map the sample onto the latent space of the normal heartbeat. As the reconstruction of the ECG signal is based only on normal characteristics of the cardiac cycle, therefore higher reconstruction loss is supposed to be incurred and reconstruction quality of anomaly ECG class would be low as the model gathered the reconstruction-experience only from normal ECG samples. On the contrary, the reconstruction loss of the normal test ECG signal is expected to be low as it has been reconstructed from the latent space representation formed from the normal ECG training samples only.

The complete procedure of the proposed *ECG-NET* is summarized in Algorithm 1.

5. Experimental evaluation

Below we provide the details of the ECG dataset used followed by the data analysis, tensor creation, training, testing and validation datasets generation, followed by compared methods, operating environment and hyper-parameters setting, as well as performance evaluation measures for our experimental evaluation. Source code of the proposed model will be available on request for non commercial use only.

Algorithm 1 : Proposed ECG-NET

Input: Training set (normal ECG samples), validation set (normal ECG samples) and test set (both normal and anomaly ECG samples).

Output: Predicted test ECG as normal or anomaly.

Hyper-parameters: Activation function: ReLU, number of epoch: 200, learning rate: 0.001, optimizer: adaptive momentum L1 loss and dropout: 20%.

ECG-NET (Hyper-parameters values):

```

1: BEGIN
  /*Training Phase*/
2: for loop in range (normal ECG training set) do
3:   Pass the samples into the 3-layered LSTM encoder block in order
   to generate the latent space representation of the ECG signal.
4:   Pass the encoded ECG to 3-layered LSTM decoder block and
   reconstruct the ECG signal.
5:   Compute the reconstruction loss.
6:   Store and plot the reconstruction loss.
7: end for
8: Pass the validation normal ECG signals to 3- layered encoder fol-
   lowed by 3-layered decoder block and compute the reconstruction
   loss.
  /*Testing Phase*/
9: for loop in range (normal and anomaly ECG test set) do
10:  Pass the ECG test sample into the 3-layered decoder block of the
   trained LSTM autoencoder model to reconstruct the test ECG.
11:  Calculate the reconstruction loss of the test ECG sample.
12:  Select the reconstruction loss threshold value  $\tau$  using manual or
   Kapur's thresholding method to predict ECG type (normal and
   anomaly).
13:  if reconstruction loss  $\leq \tau$  then
14:    Predict the test ECG as normal .
15:  else
16:    Predict the test ECG as anomaly.
17:  end if
18: end for
19: END

```

5.1. ECG dataset used

In this work, we have used publicly available *ECG5000*² dataset. The original dataset was released by Eamonn Keogh and Yanping Chen that is downloaded from physionet. This database is a 20 hour long BIDMC congestive Heart Failure Database (chfdb) ECG database reported as “chf07” which is publicly available in the UCR Time Series Classification archive (Bagnall et al.) (Bagnall et al., 2018) 2018.

The available *ECG5000* dataset is a preprocessed dataset, where the data preprocessing was performed in two steps, viz., the first step extracts each heartbeat and second step makes each heartbeat of equal length using interpolation. 5000 heartbeats were randomly selected for making the *ECG5000* dataset. The dataset comprises both normal patients' ECG as well as the ECG of those patients who have severe congestive heart failure.

The *ECG5000* dataset contains real world time sequences ECG heartbeat data. There are 5000 ECG time series signals in this database. The total length of the each ECG is 140. Therefore, the entire sequence i.e., (1×140) is depicted as a single ECG report of a single patient, and the dimension of *ECG5000* is (5000×141) . Among 141 columns of the *ECG5000*, the first 140 columns represent the single ECG report, and the last column i.e., the 141th column contains the class of that specific

ECG signal. The ECG heartbeat is categorized into five categories or classes. The five categorized ECG samples are of types Normal (N), R-on-T Premature Ventricular Contraction (R-on-T PVC), Premature Ventricular Contraction (PVC), Supra-ventricular Premature or Ectopic Beat (SP or EB) and Unclassified Beat (UB) that having 2919, 1767, 96, 194 and 24 samples respectively as depicted in Fig. 7.

5.1.1. Data analysis

In data analysis phase, we analyzed the morphological variability graph for each class. As we know, a normal heartbeat shows a rhythmic pattern, whereas, abnormal heartbeats merely contains any rhythmic patterns in it. We have explored both the normal and abnormal ECGs separately to find out whether any dissimilarity present among all types of ECG signals. Another intuition for exploration was to find out whether different type of abnormalities follows somehow similar pattern among them or not. Going through the graphical analysis of means and standard deviation of each class, it is clearly seen that the time series graph of the normal class follows quite different pattern from the anomaly classes. The morphological variability graph of each ECG class is shown in Fig. 8. The highlighted regions indicate the overall average amplitude (dark blue line) surrounded by one standard deviation, in both sides (light blue). As all the anomaly classes (viz., PVC, R-on-T, SP, UB) follow a similar pattern of heart rhythm, it is easier to differentiate between normal and anomaly ECG. This data analysis is done to understand and to study the difference between normal and anomaly classes of ECG signal.

5.1.2. Tensor creation

Tensors have been created for each five classes which are needed for to use them further for training and testing purpose. So, we created 5 types of tensors corresponding to Normal, PVC, R-on-T, SP and UB ECG types. Thus each ECG signal tensor is depicted by a feature vector of length 140.

5.1.3. Training, testing and validation dataset

We have performed experiments with two different kinds of data splitting strategy as discussed below.

In the first find of splitting, 80% of the entire normal ECG dataset (i.e., $2101 + 234 = 2335$) is used to train and validate the model whereas remaining 20% (i.e., 584) of the total normal ECG samples along with 100% (i.e., 1767, 96, 194, 24 respectively for PVC, R-on-T, SP and UB, total 2081) anomaly ECG samples are utilized for testing purpose. The 80% training and validation dataset is further decomposed into 90% (i.e., 2101) and 10% (i.e., 234) respectively for training and validation purpose.

Whereas, in the second kind of splitting we have used 292 normal ECG samples as training plus validation (in which 10% of 292 are kept in validation set) and 100% of the test samples i.e., 4500 (containing 2627 normal and 1873 anomaly) ECG signals are used to test the model.

Please note that, our proposed LSTM based autoencoder model is trained only on training data such that the encoder block of the autoencoder can produce the latent space representation of the normal ECG signals which is further used by the decoder block to reconstruct the test ECG samples. Once the test ECG sample is normal the reconstruction loss would be low, whereas the reconstruction loss would be high if the test sample is of anomaly type. The summary of ECG signals present in training, validation and testing sets is shown in Table 1.

5.2. Methods compared

The effectiveness of the proposed method is tested by comparing it with three other recently published methods mentioned in the literature viz., Dutta et al. (2021), Noor et al. (2021) and Pereira and Silveira (2019). To test the effectiveness of our proposed algorithm, it is necessary to compare it with other existing algorithms which are similar in nature and have used the same dataset. All the three works carried

² <http://www.timeseriesclassification.com/description.php?Dataset=ECG5000>.

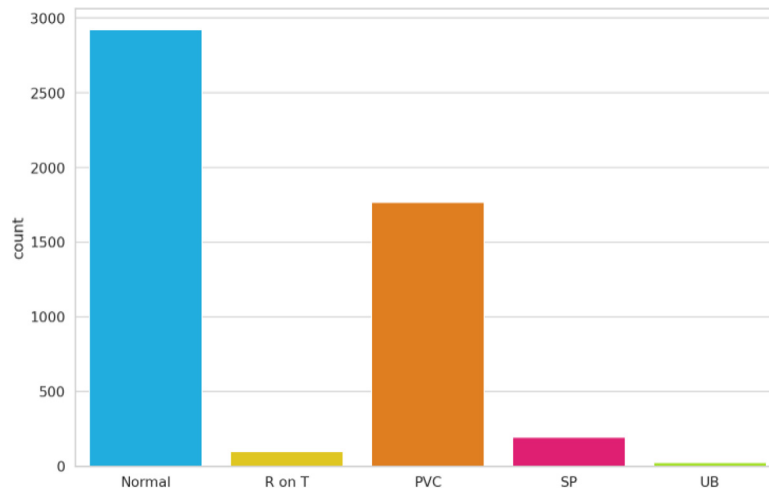


Fig. 7. Five types of heartbeat classes present in ECG5000 dataset.

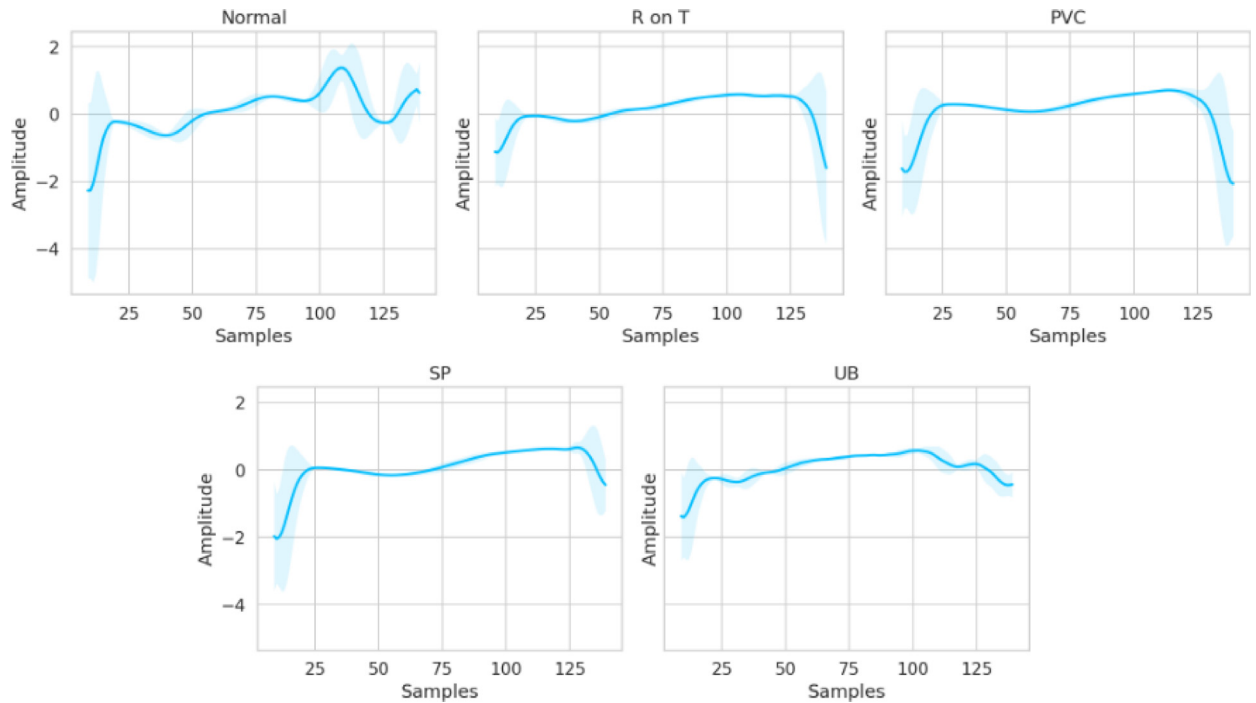


Fig. 8. Morphological variability graph of each ECG class. The highlighted regions indicate the overall average amplitude (dark blue line) surrounded by standard deviation, in both sides (light blue).

out by the aforesaid researchers (Dutta et al., 2021; Noor et al., 2021; Pereira and Silveira, 2019) are done on the same ECG5000 dataset considering the problem as binary classification task viz., normal class and anomaly class.

The performance of the proposed method is also compared with Murat et al. (2020) work with ECG5000 data sets. Further, for the sake of comparison Vision transformer (ViT) based method (Dosovitskiy et al., 2021) is also implemented by converting the ECG5000 dataset as two dimensional image for each ECG sample, representing amplitude value in the vertical axis against 142 time points in horizontal axis and then dividing the images into 32 small regions known as patches having dimension 8×13 each.

It is worth noting that, Dutta et al. (2021) also used autoencoder based LSTM model for ECG anomaly prediction. Relatively small architecture (as compared with the proposed method) have been proposed by the authors comprising only two LSTM Layers in both encoder

and decoder block. Noor et al. (2021) proposed another approach by combining RNN with LSTM autoencoder. However no dropout method was applied in both encoder and decoder block to address overfitting and underfitting issues.

Pereira and Silveira (2019) used variational autoencoder with SVM classifier, cluster and wasserstein distance similarity to predict ECG anomaly. Authors have used traditional machine learning approach along with deep learning technique for prediction.

Differences between the proposed and the existing compared techniques (Dutta et al., 2021; Noor et al., 2021; Pereira and Silveira, 2019) are highlighted in Section 2.1.

5.3. Operating environment and hyper-parameters setting

The following subsections describe the operating environment along with hyperparameters setting.

Table 1
Training, validation and testing ECG dataset.

Data splitting	Dataset	No. of ECG samples
First kind of splitting	Training (Normal ECG)	2101
	Validation (Normal ECG)	234
	Test (Normal ECG)	584
	Test (Anomaly ECG)	
	PVC	1767
	R-on-T	96
	SP	194
Second kind of splitting	UB	24
	Total(Anomaly)	2081
	Training (Normal ECG)	263
	Validation (Normal ECG)	29
	Test (Normal ECG)	2627
	Test (Anomaly ECG)	
	PVC	1590
	R-on-T	86
	SP	175
	UB	22
	Total(Anomaly)	1873

5.3.1. Operating environment

The experiments are performed in Google Colab Pro Cloud platform using GPU environment. The implementation of the proposed algorithm is done in Pytorch 1.11.0+cu113. The numpy 1.21.6 version and pandas 1.3.5 version are used for the implementation.

5.3.2. Hyper-parameters settings of the proposed architecture

The overall hyper-parameters of the proposed method are empirically determined keeping in view the following points:

The dropout rate (Goodfellow et al., 2016) in a neural network determines the fraction of neurons to be dropped during training. During the training process, it has been observed in our experiments that by setting the dropout rate as 20%, the model is able to capture more desired patterns from the normal ECG samples.

Choosing the appropriate activation function for a specific type of problem is very necessary. The proposed method uses ReLU as activation function as it is computationally effective, and is able to introduce non-linearity to the network model. ReLU is also able to speed up the convergence of the neural network during training as well as avoid vanishing gradient problems (Goodfellow et al., 2016).

Adaptive Momentum estimation or ADAM optimizer has been used in the proposed architecture. Due to its proficiency with sparse gradients and its convergence characteristics, the Adam optimizer fits for the proposed architecture (Goodfellow et al., 2016).

Mean absolute error (MAE) or L1 loss is frequently employed in deep learning models when sparse outputs are desired. The previous works (Dutta et al., 2021; Noor et al., 2021) related to anomaly detection in ECGs by different deep learning models have used L1 loss as loss function. Keeping in view, we also have used L1 loss in the proposed architecture.

Here, we have chosen the batch size as 1 because we want to train our model with the single ECG signal at a time. While the model is supplied with a single ECG at a time, it will try to update the corresponding weights and biases to better fit each individual ECG signal during the training process. It is allowing the model to capture the distinctive features of each ECG signal which indeed lead to accuracy improvement.

The learning rate plays an important role while the optimization algorithm is searching for the minimum of loss function. Empirically it has been observed that choosing the learning rate as 0.001 works well for several deep learning models. One key reason is that, if the learning rate becomes too large and too small then it may lead to divergent optimization or slow convergence respectively.

Choosing the appropriate number of epoch (or maximum epoch) is significantly important in deep learning-based models as it instructs the model how many times it should iterate through the whole training dataset. However, setting the maximum epoch before training a model

Table 2

Overview of training hyper-parameters of ECG-NET model.

Batch size	1
Activation function	ReLU
No. of epoch	200 (for first data splitting) 600 (for second data splitting)
Dropout rate	20%
Learning rate	0.001
Optimizer	Adaptive momentum
Loss	L1 Loss

could be a crucial factor in avoiding overfitting or underfitting. Theoretically, for finding the optimal (or maximum) epoch is to monitor the model's performance on a validation set over the epoch and stop training when the performance on the validation set reaches to plateau or starts to increase, which is a sign of overfitting. The usual practice to set the optimum epoch is to experiment with different epoch values and see the model's performance on validation data and empirically choose the optimum threshold value for which the validation loss is minimum. In our proposed approach also we have adopted the same concept and empirically set the epoch value for which the validation loss becomes minimum.

The empirically determined hyper-parameters used in the proposed architecture are summarized in Table 2.

5.4. Performance evaluation measures

Here, we have used four different kinds of performance evaluation measures viz., percentage accuracy, precision, recall, micro-averaged F1 measure for evaluating the overall performance of our proposed ECG-NET method.

5.4.1. Percentage accuracy

Here, we have reported the results on test case accuracy only, that is percentage of correctly classified test ECG signals out of total test ECG signals which is defined as

$$\text{percentage accuracy} = \frac{\# \text{ test ECG signal correctly classified}}{\text{Total number of test ECG signals}} \times 100 \quad (10)$$

5.4.2. Precision

Precision is the ratio of number of test ECG signals correctly classified into i th class to the total number of test ECG signals classified into i th class. The precision (p_i) of i th class and overall precision are defined respectively, as

$$\text{precision}_i(p_i) = \frac{\# \text{ test ECG signals correctly classified into } i\text{th class}}{\# \text{ test ECG signals classified into } i\text{th class}} \quad (11)$$

$$\text{Overall precision} = \frac{1}{N} \sum_{i=1}^N p_i \quad (12)$$

where, N is the number of class present in the dataset.

5.4.3. Recall

Recall is the ratio of number of test ECG signals correctly classified into i th class to the total number of test ECG signals that are truly present in i th class. The recall (r_i) of the i th class and overall recall are defined respectively, as

$$\text{recall}_i(r_i) = \frac{\# \text{ test ECG signals correctly classified into } i\text{th class}}{\# \text{ test ECG signals that are truly present into } i\text{th class}} \quad (13)$$

$$\text{Overall recall} = \frac{1}{N} \sum_{i=1}^N r_i \quad (14)$$

where, N is the number of class present in the ECG dataset.

5.4.4. Micro-averaged (F_1) measure

Micro-averaged F_1 measure is computed by first creating a global contingency table whose cell values are the sum of the corresponding cells in the per-class contingency tables. Then use this global contingency table to compute the micro-averaged performance scores. Micro-averaged F_1 gives equal weightage on each sample. Micro-averaged F_1 is defined as

$$\text{Micro-averaged } F_1 = \frac{2 \times \frac{1}{N} \sum_{i=1}^N p_i \times \frac{1}{N} \sum_{i=1}^N r_i}{\frac{1}{N} \sum_{i=1}^N p_i + \frac{1}{N} \sum_{i=1}^N r_i} \quad (15)$$

6. Experimental results and analysis

Details of the experimental outcomes obtained by the proposed ECG-NET method using first kind of data splitting (viz., 80% of normal ECGs as training dataset and remaining 20% of normal ECGs as well as 100% of anomaly ECGs for testing dataset) are elaborated in this section below. Additionally, the experimental results of the proposed method using second kind of splitting are reported in the supplementary material of this article.

6.1. Experimental results on training and validation data

The proposed method employed two threshold selection policies in the testing phase. Therefore, it is worthy to mention that the training phase of the proposed method is same for both the cases. The proposed model has been trained over 200 epochs with learning rate 0.001. Fig. 9 depicts the decay of the L1 loss over epochs for training and validation data sets. It can be clearly observed from the graph that over the epoch both the training loss and validation loss are continuously decreasing and finally saturate after around 200 epochs.

As mentioned earlier in Section 4, we have trained our model only on normal ECG samples to learn the latent space representation of the normal ECG samples. Based on this latent space representation of the normal ECG samples, the information obtained by the model is further used in the testing phase for both normal and abnormal ECG classification. To validate the aforementioned idea it is important to observe the relationship between the reconstruction loss and its corresponding frequency distribution values, after the entire training phase is over. Plotting these obtained values as histogram helps us to observe the said relationship. Fig. 10 indicates that, most of the reconstruction loss of the training (normal) ECG samples are covered when the reconstruction loss value is under 23.

After training the model, 10% (i.e., 234) of normal ECG samples are passed as validation data to observe the variance of waveform between the original and reconstructed ECG samples. In Fig. 11, it is clearly seen that the waveform of original and reconstructed ECG signals are quite identical. Fig. 11 depicts the original normal ECG signals (as validation ECG samples) along with their corresponding reconstructed ECG signals of four representative ECGs, blue line indicates the original validation ECG signal whereas yellow line represents the reconstructed signal of the same ECG used for the validation purpose.

6.2. Experimental results on test data

The test dataset contains 20% of the normal ECG samples (i.e., 584 normal ECG signals) and all the anomaly ECG samples (i.e., 2081 ECG signals) available on the ECG5000 dataset. Following subsections describe the testing process through manual threshold selection as well as automated threshold selection of the proposed ECG-NET approach.

6.2.1. Results of testing process of ECG-NET model

As mentioned earlier, the proposed model has been trained only on the normal ECG signals to learn the latent space representation. Hence, in the testing phase, the test ECG signals (both normal and anomaly test samples), are fed into the model and the reconstruction loss of each ECG test samples are calculated. Then the normalized frequency of the reconstruction loss of the anomaly ECG samples as well as normal test ECG samples have been plotted with respect to their corresponding reconstruction loss values, as shown in Fig. 12. From the plot we observe that, that the reconstruction loss value of anomaly ECG test samples yield larger reconstruction loss value in comparison with normal ECG test samples. The blue curve of the graph represents the reconstruction loss behavior of normal ECG test signals, whereas the reconstruction loss behavior of abnormal test ECG samples are denoted by yellow curve. The valley formed by plotting the reconstruction loss of both normal as well as abnormal ECG samples, strongly suggests that an optimum threshold value can differentiate the behavior of normal and abnormal ECG samples for the prediction process. Therefore we employed two reconstruction loss threshold selection technique viz., one using manual thresholding technique by visually observing the valley of the curve, and another using Kapur's thresholding (Kapur et al., 1985) method.

Testing process of manual threshold selection procedure: By inspecting the frequency curve (histogram plot) of the reconstruction losses we manually determined the threshold value to be 23 (shown in red color dotted line) for which the curve attains the valley portion suggesting the fact that the test ECG sample with reconstruction loss value less than or equal to 23 would be predicted as normal and above this threshold would be determined as anomaly ECG signal.

After placing the reconstruction losses threshold value at 23, it is clearly seen from Fig. 12, that most of the normal ECG data losses (shown in blue color region in the curve) fall under the said threshold value, whereas most of the anomaly ECG (shown in yellow color region in the curve) losses lie above the specified threshold.

Testing process of automated threshold selection procedure: In ECG5000 dataset, we have considered all 2081 anomaly ECG samples as well as 584 normal ECG samples for testing the proposed model. In real life scenario, when the dataset is too large, manually examining the frequency curve (histogram plot) of the reconstruction losses to choose the threshold value may not be feasible as it entails time-consuming human involvement. To handle such scenario, automated threshold selection procedure can be applied. Therefore, as mentioned earlier, the proposed approach also uses an automated threshold selection procedure inspired by Kapur's thresholding method (Kapur et al., 1985), (described in Section 4.3.2). In Fig. 12, the dotted line suggests the optimum threshold value to be at 23 obtained by applying the Kapur's method on the reconstruction losses of ECG test samples.

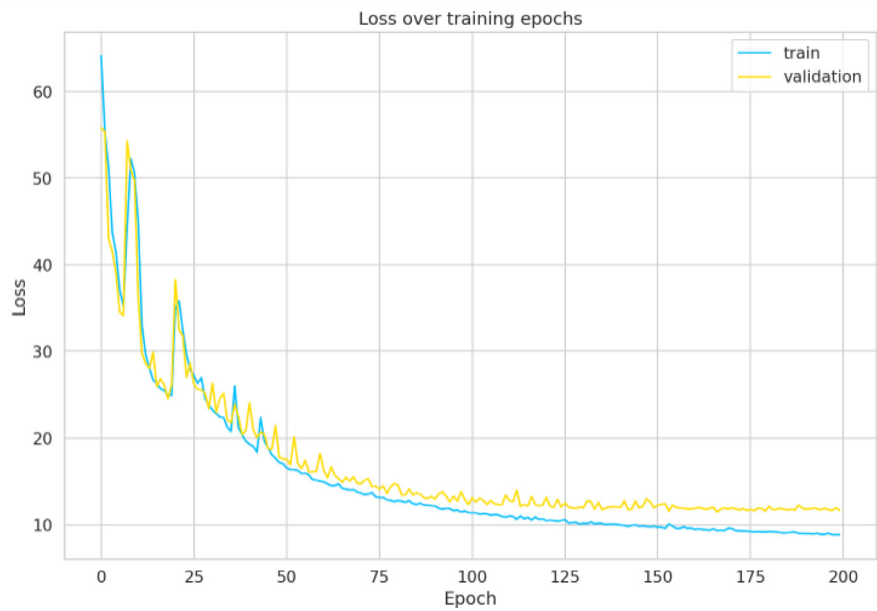


Fig. 9. L1 training and validation losses over the epochs.

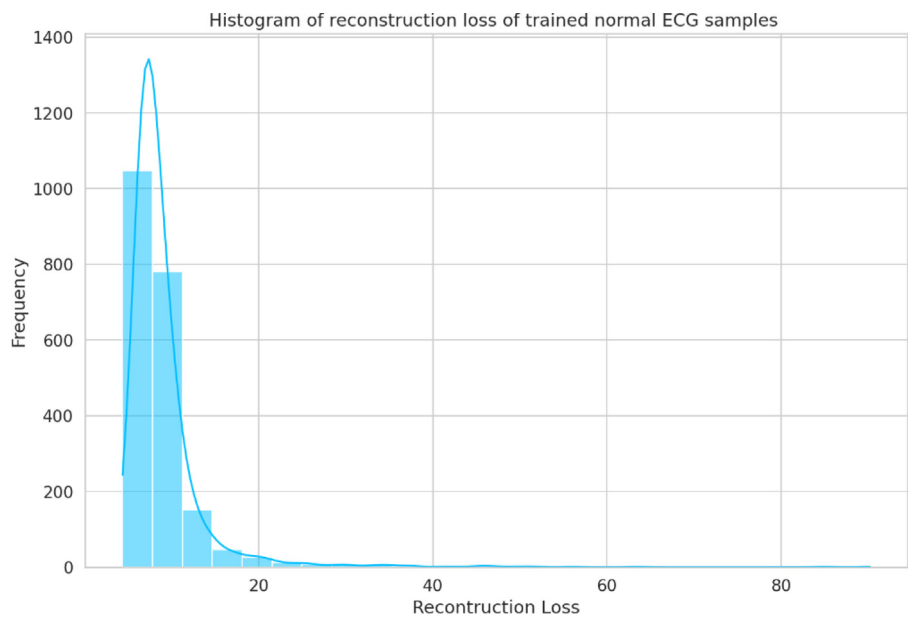


Fig. 10. Reconstruction loss graph of normal ECG samples after training phase.

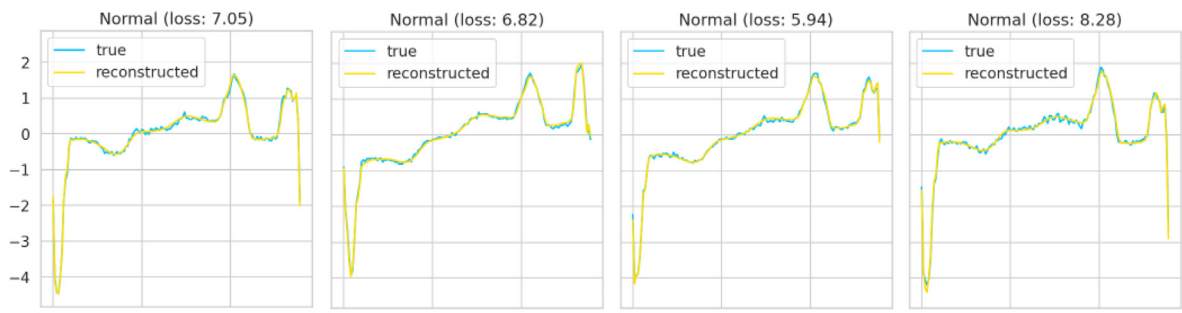


Fig. 11. Reconstruction of the validation normal ECG signals obtained using proposed method.

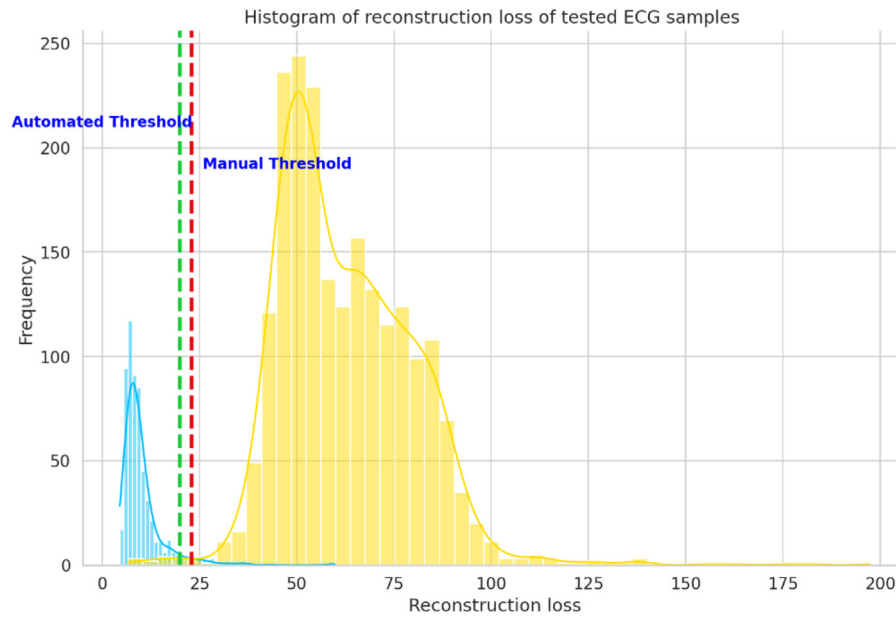


Fig. 12. The reconstruction loss graph with respect to histogram of ECG test dataset along with manual and automated threshold.

Table 3

Details of correctly and incorrectly classified ECG test samples using manual thresholding procedure manual thresholding procedure adopted in the proposed *ECG-NET*.

Dataset	Types of ECG	# Total	# Samples detected as Normal	# Samples detected as Anomaly
Normal test data	Normal ECG	584	562	22
Anomaly dataset	PVC	1767	1	1766
	R-on-T	96	8	88
	SP	194	2	192
	UB	24	4	20

Table 4

Confusion matrix of the classified test ECG samples by manual thresholding procedure adopted in the proposed *ECG-NET*.

Manual thresholding		True class	
		Normal	Anomaly
Predicted class	Normal	562	22
	Anomaly	13	2068

Table 5

Overview performances of *ECG-NET* model using manual thresholding procedure on *ECG5000* dataset.

Accuracy	98.68%
Precision	0.9623
Recall	0.9774
F1 score	0.9698

To observe the closeness of automated threshold value vs. manual threshold value getting from the test dataset side-by-side, we have plotted both threshold lines in Fig. 12, red dotted line denotes threshold valued curated by manual process, whereas automated threshold values is denoted by green line in the graph.

6.2.2 Reconstruction of the test ECG sample

As stated earlier, the reconstruction loss values of normal and anomaly ECG samples play an important role in the proposed architecture for classification of normal and anomaly ECGs. Figs. 13

and 14 respectively depicts the normal and anomaly (original) ECG signals along with their corresponding reconstructed ECG signals of six representative ECGs.

The original and reconstructed signals are represented by blue and yellow line respectively in the Figures. Reconstruction loss for each sample is also mentioned in the respective graph. Minute observation of reconstruction loss curves reveals that normal ECG samples have very low reconstruction loss values compared to reconstruction loss values of the anomaly ECG samples. This is also evident from Fig. 13 that the reconstructed normal ECG signals and the corresponding original normal ECG signals are coinciding each other. Which indicates that the proposed model is capable of capturing the true behavior of the normal ECG samples. In case of anomaly ECG samples, reconstruction curve of each anomaly sample deviates from the true anomaly ECG sample resulting higher reconstruction loss value. This is because of the fact that the model is being trained only using normal characteristics of the cardiac cycle, therefore the latent space representation generated by the encoder block of the proposed model is based on the normal ECG samples only, hence unable to capture the true behavior of the anomaly ECG sample through the reconstruction procedure. This behavior (of low reconstruction loss values for normal ECG sample and high loss values for anomaly ECG signal) is being exploited in the proposed method to demarcate/classify the normal ECG signals from the anomaly class by finding out the suitable threshold value of the reconstruction loss from the frequency distribution of the reconstruction loss curve (as detailed in Section 4.3).

6.2.3 Numerical results of the proposed *ECG-NET* with manual threshold selection

The experimental results in terms of correctly classified normal and anomaly ECG samples obtained using *ECG-NET* with manual thresholding procedure adopted in the proposed method, are summarized in Table 3.

From the Table 3 it can be clearly seen that out of total 584 normal ECG signals, the proposed method has correctly predicted 562 normal ECG signals; whereas, among 2081 anomaly ECG samples, the proposed method able to correctly classify 2068 samples as anomaly ECG. Result in terms of confusion matrix for normal and anomaly ECG signals using manual thresholding procedure are shown in Table 4. The proposed method achieved overall 98.68% accuracy with precision,

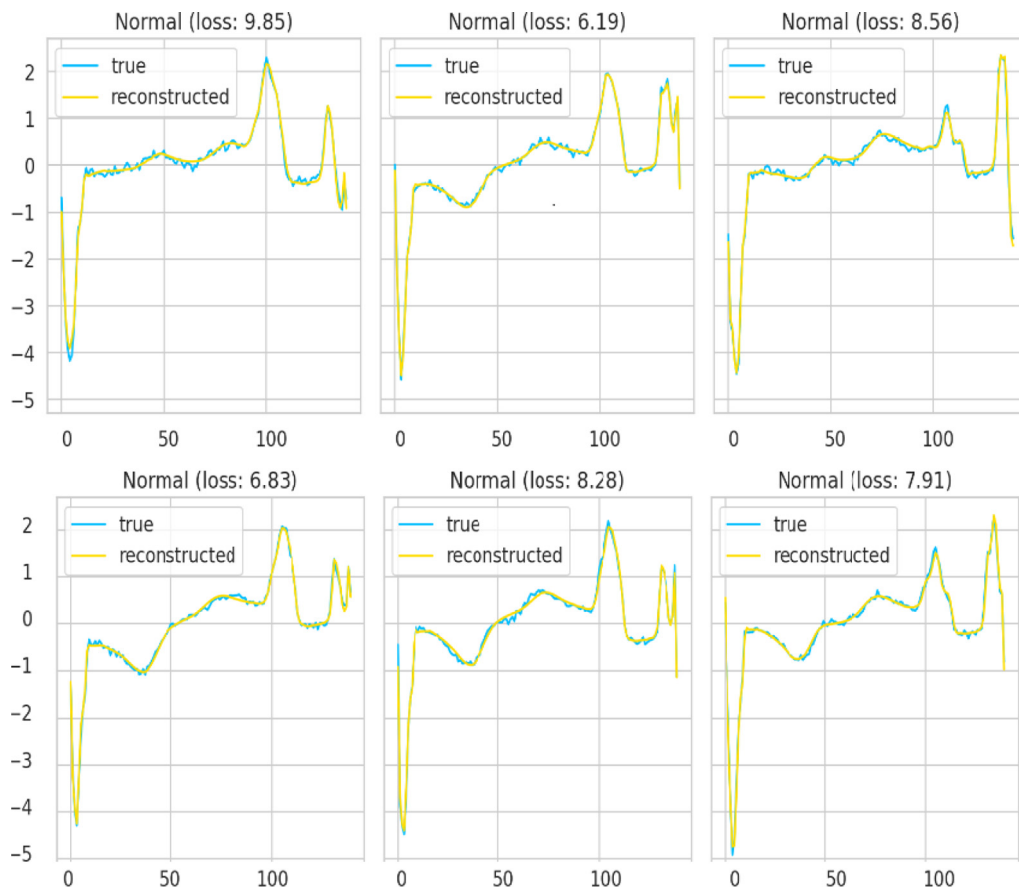


Fig. 13. Reconstruction of the normal ECG signal obtained using proposed method.

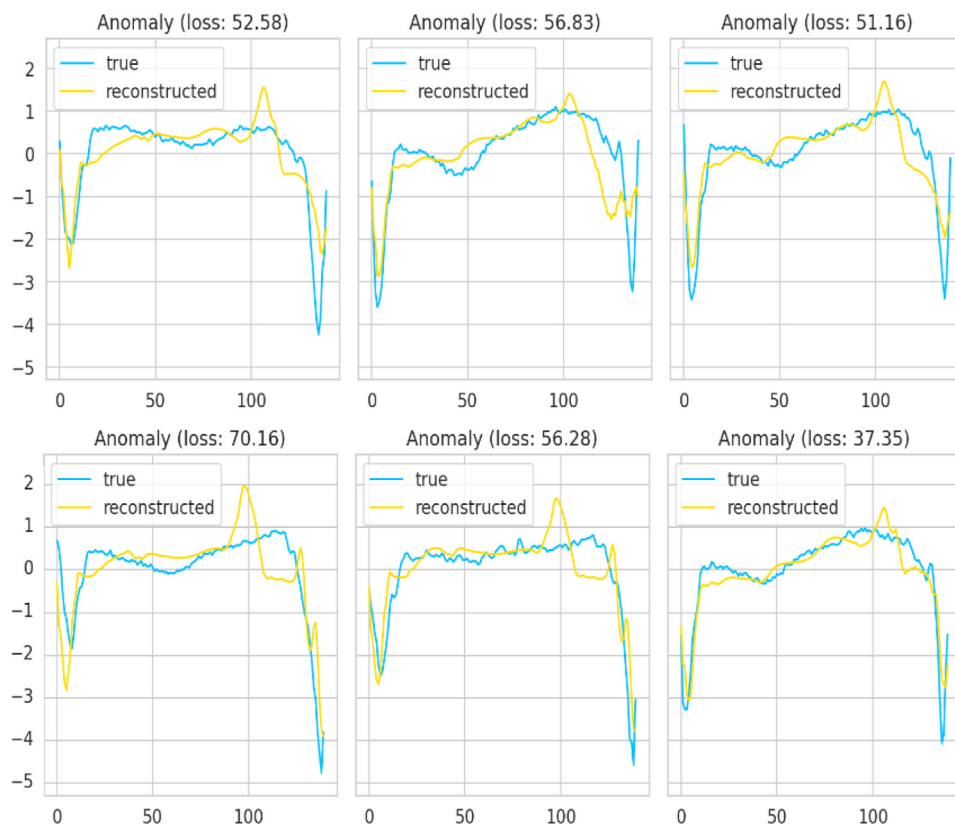
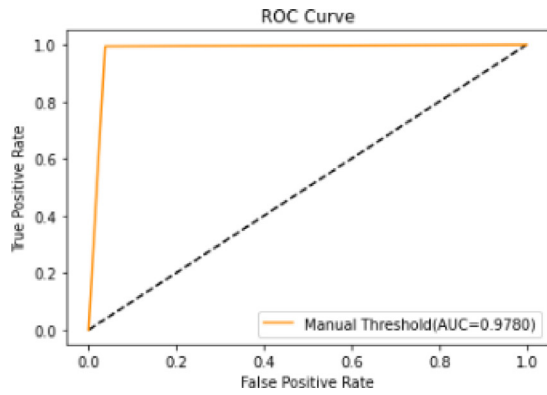
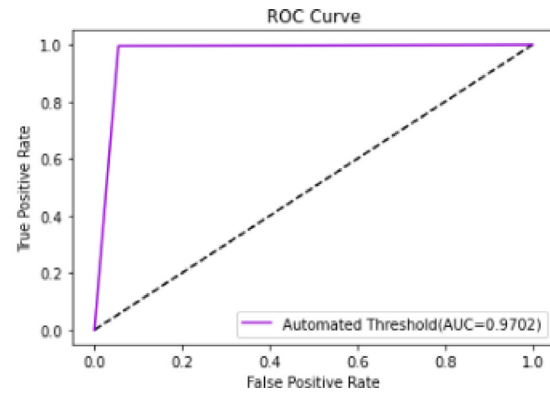


Fig. 14. Reconstruction of the anomaly ECG signal obtained using proposed method.



(a) Manual Thresholding procedure(training normal ECG = 2101, test normal and anomaly = 2665)



(b) Automated Thresholding procedure(training normal ECG = 2101, test normal and anomaly = 2665)

Fig. 15. ROC curve and AUC for manual and automated thresholding procedure.

Table 6

Details of correctly and incorrectly classified ECG test samples using automated threshold procedure adopted in the proposed ECG-NET.

Dataset	Types of ECG	Total No. of sample	Detected as normal	Detected as anomaly
Normal test data	Normal ECG	584	552	32
Anomaly dataset	PVC	1767	0	1767
	R-on-T	96	6	90
	SP	194	0	194
	UB	24	4	20

Table 7

Confusion matrix of the classified test ECG samples by automated thresholding procedure.

		True class	
		Normal	Anomaly
Predicted class	Normal	552	32
	Anomaly	10	2071

recall and $F1$ values as 0.9623, 0.9774 and 0.9698, respectively which are summarized in Table 5.

6.2.4 Numerical results of the proposed ECG-NET with automated threshold selection

Quantitative classification results of normal and abnormal ECG samples obtained by automated thresholding (using Kapur's) method are summarized in Table 6. From Table 6, we can clearly see that the proposed method correctly predicted 552 normal ECG signals out of a total of 584 normal ECG signals. Whereas, out of 2081 abnormal ECG samples, the proposed automated thresholding method can correctly classify 2071 samples as abnormal ECG signals. Table 7 summarizes the confusion matrix of the normal and anomaly ECG classification outcome using automated thresholding technique adopted in the proposed method.

This automated thresholding based approach achieves 98.42% overall accuracy. Out of total 584 normal ECG signals the proposed method has correctly predicted 552 normal ECG signals whereas, out of 2081 anomaly samples, it predicted 2071 samples correctly. The overall performance in terms of Accuracy, Precision, Recall, $F1$ score of the proposed automated thresholding method are summarized in Table 8.

Table 8

Performance overview of ECG-NET model using automated thresholding procedure on ECG5000 dataset.

Accuracy	98.42%
Precision	0.9452
Recall	0.9822
$F1$ score	0.9633

6.2.5 Receiver operating characteristic curve

Here, we measure the performance of our proposed model through receiver operating characteristic curve (ROC) that depicts how well a classification model performs across all classification thresholds. In this subsection, the ROC curve and AUC score of our proposed model are shown in Fig. 15.

In Fig. 15(a) and (b) show the ROC curve and AUC score of manual and automated threshold selection procedure of our proposed method ECG-NET with 2101 numbers of training normal ECG samples and 2665 numbers of test ECG samples respectively.

6.3 Comparison between related works and our proposed ECG-NET model

The comparative results in terms of classification accuracy obtained by our proposed ECG-NET approach with the results reported in the related existing works (Murat et al., 2020; Dutta et al., 2021; Noor et al., 2021; Pereira and Silveira, 2019) from the literature and implemented ViT based method (Dosovitskiy et al., 2021) are summarized in Table 9. From the summarized results it can be observed that proposed ECG-NET method with manual thresholding technique achieved the highest accuracy (98.68%, 98.56%) among all other compared methods. Whereas the proposed ECG-NET method with automated thresholding technique (with accuracy values 98.42%, 98.36%) marginally lagging behind the VRAE+SVM method (98.43%). However, VRAE+SVM method requires normal and abnormal samples both in training phase of the

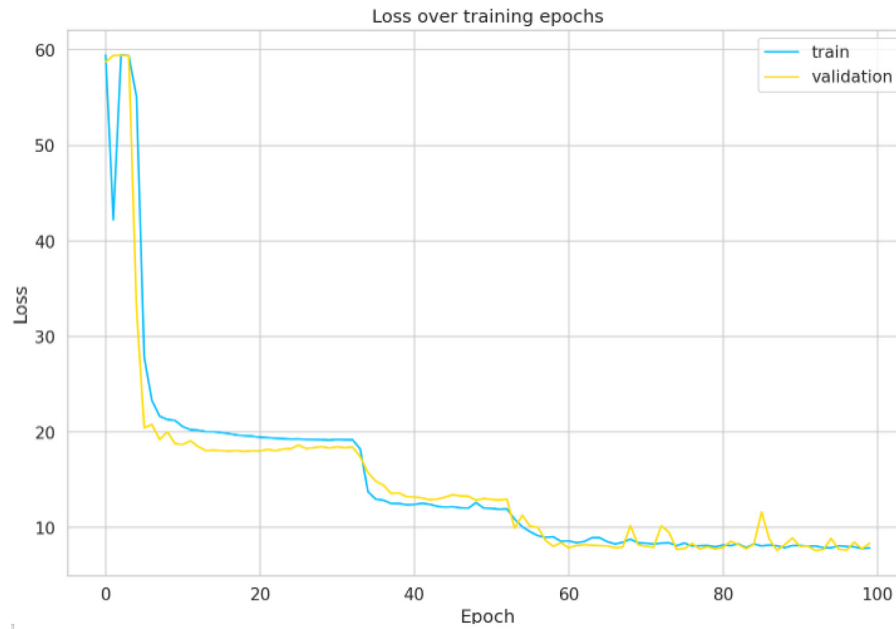


Fig. 16. L1 training and validation losses over the epochs.

Table 9

Performance comparison in terms of percentage accuracy of *ECG-NET* and other existing methods with various data splitting.

Method	Model	Splitting	Accuracy
Pereira and Silveira (2019)	VRAE + SVM	#training normal & anomaly ECGs = 400 #validation normal & anomaly ECGs = 100 #test normal & anomaly ECGs = 4500	98.43%
Pereira and Silveira (2019)	VRAE + Wasserstein	#training normal & anomaly ECGs = 400 #validation normal & anomaly ECGs = 100 #test normal & anomaly ECGs = 4500	95.96%
Pereira and Silveira (2019)	VRAE + Clustering	#training normal & anomaly ECGs = 400 #validation normal & anomaly ECGs = 100 #test normal & anomaly ECGs = 4500	95.96%
Dutta et al. (2021)	LSTM Auto-encoder	#training normal & anomaly ECGs = 500 #test normal & anomaly = 4500	97.93%
Noor et al. (2021)	RNN-LSTM Auto-encoder	#training normal ECGs only = 1401 #validation normal ECGs only = 963 #test normal & anomaly = 438	97.24%
Murat et al. (2020)	CNN-LSTM	#training normal and anomaly ECGs = 4000 #validation Normal & anomaly ECGs = 500 #test normal & anomaly = 500	93.90%
Dosovitskiy et al. (2021)	ViT	#training normal & anomaly ECGs = 3500 #validation normal & Anomaly ECGs = 500 # test normal & anomaly = 1000	97.67%
Proposed ECG-NET	ECG-NET with manual threshold	# training normal ECGs only = 2101 #validation normal ECGs only = 234 # test normal & anomaly = 2665	98.68%
	ECG-NET with automated threshold	#training normal ECGs only = 2101 #validation normal ECGs only = 234 # test normal & anomaly = 2665	98.42%
	ECG-NET with manual threshold	# training normal ECGs only = 263 #validation normal ECGs only = 29 # test normal & anomaly = 4500	98.56%
	ECG-NET with automated threshold	# training normal ECGs only = 263 #validation normal ECGs only = 29 # test normal & anomaly = 4500	98.36%

model, whereas the proposed method require only normal ECG samples (without the need of anomaly signal) to train the model.

Another advantage of the proposed *ECG-NET* method is that the model can predict normal and anomalous signal from the reconstructed

Table 10Training, validation and testing split in *TwoLeadECG* dataset.

Dataset		No. of ECG samples
Training (Normal ECG)		417
Validation (Normal ECG)		47
Test	Normal ECG	117
	Anomaly ECG	581

Table 11Confusion matrix of the classified test ECG samples by manual thresholding procedure adopted in the proposed *ECG-NET*.

Manual thresholding		True class	
		Normal	Anomaly
Predicted class	Normal	114	3
	Anomaly	7	574

Table 12Confusion matrix of the classified test ECG samples by automated thresholding procedure adopted in the proposed *ECG-NET*.

Automated thresholding		True class	
		Normal	Anomaly
Predicted class	Normal	115	2
	Anomaly	9	572

histogram by manual and automated thresholding procedure. When the reconstruction loss histogram is having clear-cut valley, we can go for manual threshold selection procedure by visually observing the reconstruction loss threshold values appropriately. While the reconstruction loss histogram does not have any clear-cut valley then we can go for automated thresholding procedure viz., Kapur's method as adopted in the proposed method for finding out the optimum threshold value of the reconstruction loss without manual intervention. In general, as mentioned earlier in Section 4.3.2, the automated threshold procedure using Kapur's adopted in the proposed technique can work irrespective of the presence or absence of the valley in the reconstruction loss histogram.

Therefore, in summary, the proposed method turns out to be superior compared to most of the other existing techniques compared in terms of the classification accuracy without the need of having anomaly ECG signals to train the model and without the need of any human intervention (adopting the idea of automated reconstruction loss threshold selection procedure using Kapur's method).

7 Conclusions and future work

This article presents a LSTM based autoencoder model called *ECG-NET* for classification of normal and anomaly ECGs from time sequence heartbeat data. Encoder part of the proposed *ECG-NET* model is being trained only using normal ECG samples to learn the behavior of normal heartbeat from its encoded latent space representation, whereas the decoder part tries to reconstruct the original ECG signal from its encoded latent space representation. If the reconstruction loss value is low (less than a certain threshold value) then the corresponding ECG signal is said to be normal as the reconstructed ECG is formed by the decoder block using the latent space representation produced by encoder block of the proposed *ECG-NET* trained on normal ECG data only. On the other hand the reconstruction loss for the anomaly ECG would be high. Therefore, the loss associated in the reconstruction phase is the key factor to determine the presence or absence of the abnormality in the ECG signal. Hence, two thresholding techniques viz., manual and Kapur's automated threshold selection techniques are applied to differentiate between normal and anomaly ECGs. The proposed *ECG-NET* is evaluated on publicly available *ECG5000* dataset and its performance is compared with seven other state-of-the-art counterpart techniques from

the literature. Experimental results show that the proposed *ECG-NET* produced superior performance over almost all the compared methods with highest accuracy value 98.68%.

In future, other Recurrent Neural Network based models like Bi-LSTM, GRU, Bi-GRU etc. can also be used in the autoencoder architecture of the proposed method. Furthermore, transfer learning can also be adopted in the proposed model to use the pre-trained model on other similar ECG data sets in order to address the scarcity of the training datasets. As the reconstruction frequency distributions for normal and anomaly classes turns out to be overlapping in nature, therefore effort may also be made to find the reconstruction loss threshold value by borrowing the idea of automated fuzzy thresholding technique. Finally, robustness of the *ECG-NET* may also be tested further in some more real life ECG signals collected from different medical institutes.

CRediT authorship contribution statement

Moumita Roy: Carried out overall implementation, Experiments, Simulation, Comparison, Initial drafting of the manuscript. **Sukanta Majumder:** Carried out implementation, Experiments, Comparison, Result analysis, Drafting of the manuscript. **Anindya Halder:** Developed the overall idea of the work, Data and result analysis, Manuscript revision, Drafting of the manuscript and addressed reviewers' comments. **Utpal Biswas:** Monitor the overall progress of the work, Finalized the manuscript in revised form.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Authors in this article have used publicly available dataset

Acknowledgment

Authors are thankful to the Editor/Associate Editor and anonymous reviewers for their constructive comments to improve the manuscript.

Appendix A. Additional results

In order to test the robustness, proposed *ECG-NET* method is applied on additional ECG dataset. The details of the data and results are provided below.

A.1. Additional ECG dataset used

We have additionally used publicly available *TwoLeadECG*³ dataset (Dau et al., 2018). This ECG dataset was released by Eamonn Keogh that is downloaded from physionet. Specifically, the ECG information is taken from record ltdb/15814 of the MIT-BIH Long-Term ECG database (ltdb).

The total length of the each ECG is 82. Therefore, the entire sequence i.e., (1 × 82) is depicted as a single ECG report of a single patient, and the dimension of the entire *TwoLeadECG* is (1162 × 83). Among 83 columns of the *TwoLeadECG*, the first 82 columns of the *TwoLeadECG* reflect the single ECG report, with the last column, i.e., the 83th column, representing the class of that particular ECG signal.

The *TwoLeadECG* dataset contains real world time sequences ECG heartbeat data. There are 1162 ECG time series signals present in this

³ <http://www.timeseriesclassification.com/description.php?Dataset=TwoLeadECG>.

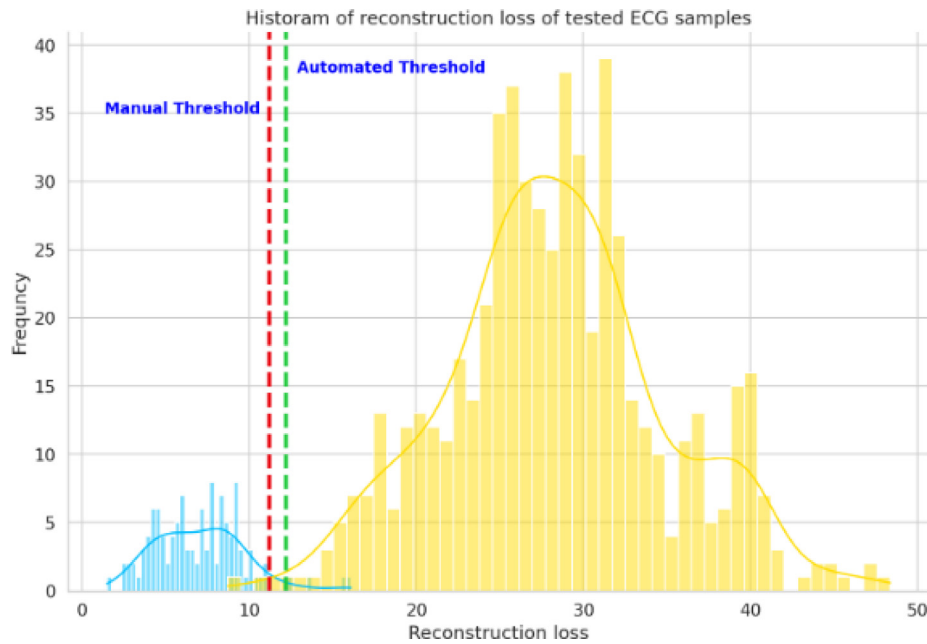


Fig. 17. The reconstruction loss graph with respect to histogram of ECG test dataset along with manual and automated threshold.

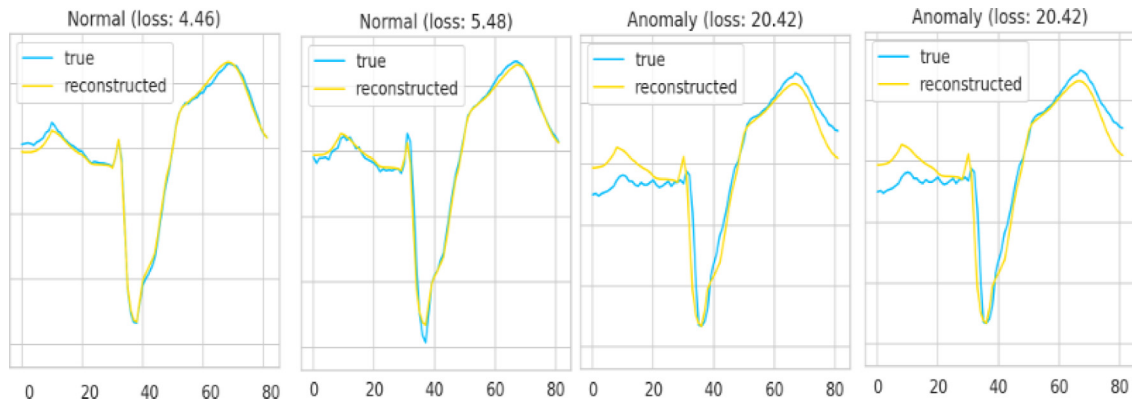


Fig. 18. Reconstruction of the normal and anomaly ECG signal obtained using proposed method.

database evenly distributed into two classes viz., normal and anomaly classes each having 581 samples.

In this work, we have split the 80% of the entire normal ECG dataset (i.e., $417 + 47 = 464$) for training and validation purpose, whereas remaining 20% (viz., 117) of the total normal ECG samples along with 100% (i.e., 581) of the anomaly ECG samples are utilized for testing purpose. The 80% training and validation dataset is further segmented into 90% (viz., 417) and 10% (i.e., 47) respectively for training and validation purpose. The summary of ECG signals present in training, validation and testing sets is shown in Table 10.

A.2. Experimental results and analysis on TwoLeadECG

In this dataset, the all operating environment and hyper-parameter are same (as those used in ECG5000 dataset) expect the number of epochs. Here, 100 epochs are used to train this model. Fig. 16 depicts the decay of the L1 loss over epochs for training and validation data sets.

After training the model, the test phase is performed. Based on this testing phase, the manual (described in Section 4.3.1) and automated threshold selection procedure (Section 4.3.2) are performed. The optimum threshold value obtained by applying the manual threshold and automated threshold using Kapur's method are 11 and 12 respectively.

In Fig. 17, the red dotted line and green dotted line suggest the optimum threshold value derived by performing the manual and automated threshold selection procedure on the histogram of ECG test samples' reconstruction loss values respectively.

The accuracy of manual and automated threshold selection procedure for TwoLeadECG dataset are 98.56% and 98.42% respectively. The confusion matrices of both manual and automated threshold selection procedure are summarized in Tables 11 and 12 respectively.

A.2.1. Reconstruction of the test ECG sample

As we previously mentioned, a key component of the proposed architecture for classifying normal and anomalous ECGs is the reconstruction loss values of normal and anomaly ECG data. In Fig. 18, the first two graph depicts the reconstruction patterns of normal ECG samples and the other remaining two shows the reconstruction patterns of anomaly ECG signals along with their corresponding reconstruction loss values.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.engappai.2023.106484>.

References

- Acar, H.A.B., Demiroz, G., Cekin, A., 1997. A supervised machine learning algorithm for arrhythmia analysis. In: *Computers in Cardiology 1997*. IEEE, pp. 433–436.
- Acharya, U.R., Oh, S.L., Hagiwara, Y., Tan, J.H., Adam, M., Gertych, A., Tan, R.S., 2017. A deep convolutional neural network model to classify heartbeats. *Comput. Biol. Med.* 89, 389–396. <http://dx.doi.org/10.1016/j.cmpb.2018.04.012>.
- Bagnall, A., Dau, H.A., Lines, J., Flynn, M., Large, J., Bostrom, A., Southam, P., Keogh, E., 2018. The UEA multivariate time series classification archive, 2018. [arXiv:1811.00075](https://arxiv.org/abs/1811.00075).
- Bajaj, K., Singh, D., Ansari, M.A., 2020. Autoencoders based deep learner for image denoising. *Procedia Comput. Sci.* 171, 1535–1541.
- Chauhan, S., Vig, L., 2015. Anomaly detection in ECG time signals via deep long short-term memory networks. *IEEE*.
- Craik, A., He, Y., Contreras-Vidal, J.L., 2019. Deep learning for electroencephalogram (EEG) classification tasks: a review. *J. Neural Eng.* 16 (3), 031001.
- Dau, H.A., Keogh, E., Kamgar, K., Yeh, C.M., Zhu, Y., Gharghabi, S., Ratanamahatana, C.A., Chen, Y., Hu, B., Begum, N., Bagnall, A., Mueen, A., Batista, G., Hexagon-ML, 2018. The UCR time series classification archive. https://www.cs.ucr.edu/~eamonn/time_series_data_2018/.
- Diker, A., Avci, D., Avci, E., Gedikpinar, M., 2019. A new technique for ECG signal classification genetic algorithm wavelet kernel extreme learning machine. *Optik* 180, 46–55. <http://dx.doi.org/10.1016/j.jllo.2018.11.065>.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N., 2021. An image is worth 16x16 words: Transformers for image recognition at scale. In: *Proceedings of International Conference on Learning Representations*. [arXiv:2010.11929](https://arxiv.org/abs/2010.11929).
- Dutta, K., Lenka, R., Nayak, S.R., Khandual, A., 2021. MED-NET: a novel approach to ECG anomaly detection using LSTM auto-encoders. *Int. J. Comput. Appl. Technol.* 65 (4), 343–357. <http://dx.doi.org/10.1504/IJCAT.2021.10040403>.
- Farhan, A., Wei, C.L., Ahmed, M.T., 2018. A qualitative overview of fuzzy logic in ECG arrhythmia classification. *Int. J. Eng. Works* 5 (11), 232–239.
- Fleming, J.S., 2012. *Interpreting the Electrocardiogram*. Springer Science & Business Media.
- Gamboa, J.C.B., 2017. Deep learning for time-series analysis. [arXiv:1701.01887](https://arxiv.org/abs/1701.01887).
- Goodfellow, I., Bengio, Y., Courville, A., 2016. *Deep Learning*. MIT Press, <http://www.deeplearningbook.org>.
- Gupta, P., Malhotra, P., Vig, L., Shroff, G., 2018. Using features from pre-trained TimeNet for clinical predictions. In: *Proceedings of International Joint Conference on Artificial Intelligence*. IJCAI, 2148, pp. 38–44. <http://dx.doi.org/10.1109/DSAA.2015.7344872>.
- Guve, D., Srivastava, A.K., Mukhopadhyay, K., Prasad, N.E., Shukla, S., Muthurajan, H., 2016. Electrocardiogram (ECG) image processing and extraction of numerical information. *Int. J. Eng. Technol. Sci. Res. (IJETSR)* 3 (7), 39–44.
- Hinton, G.E., Salakhutdinov, R.R., 2006. Reducing the dimensionality of data with neural networks. *Science* 313 (5786), 504–507.
- Hochreiter, S., 1998. The vanishing gradient problem during learning recurrent neural nets and problem solutions. *Int. J. Uncertain. Fuzziness Knowl.-Based Syst.* 6 (2), 107–116.
- Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. *Neural Comput.* 9 (8), 1735–1780.
- Ji, Z., Gong, J., Feng, J., 2021. A novel deep learning approach for anomaly detection of time series data. *Sci. Program.* 2021, 11. <http://dx.doi.org/10.1155/2021/6636270>.
- Kapur, J.N., Sahoo, P.K., Wong, A.K.C., 1985. A new method for gray-level picture thresholding using the entropy of the histogram. *Comput. Vis. Graph. Image Process.* 29 (3), 273–285.
- Khan, W., Ghazanfar, M.A., Azam, M.A., Karami, A., Alyoubi, K.H., Alfakeeh, A.S., 2022. Stock market prediction using machine learning classifiers and social media, news. *J. Ambient Intell. Humaniz. Comput.* 13, 3433–3456.
- Lynn, H.M., Kim, P., Pan, S.B., 2021. Data independent acquisition based bi-directional deep networks for biometric ECG authentication. *Appl. Sci.* 11 (3), 1125(19). <http://dx.doi.org/10.3390/app11031125>.
- Lynn, H.M., Pan, S.B., Kim, P., 2019. A deep bidirectional GRU network model for biometric electrocardiogram classification based on recurrent neural networks. *IEEE Access* 7, 145395–145405.
- Maggipinto, M., Masiero, C., Beghi, A., Susto, G.A., 2018. A convolutional autoencoder approach for feature extraction in virtual metrology. *Procedia Manuf.* 17, 126–133.
- Malhotra, P., Vig, L., Shroff, G., Agarwal, P., 2015. Long short term memory networks for anomaly detection in time series. In: *Proceedings of European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning*. Bruges, Belgium, pp. 89–94.
- Matias, P., Folgado, D., Gamboa, H., Carreiro, A.V., 2021. Robust anomaly detection in time series through variational AutoEncoders and a local similarity score. In: *Proceedings of the 14th International Joint Conference on Biomedical Engineering Systems and Technologies (BIOSTEC 2021)*. 4, Science and Technology Publications(SCITEPRESS), pp. 91–102. <http://dx.doi.org/10.5220/0010320500910102>.
- Murat, F., Yildirim, O., Talo, M., Baloglu, U.B., Demir, Y., Acharya, U.R., 2020. Application of deep learning techniques for heartbeats detection using ECG signals-analysis and review. *Comput. Biol. Med.* 120, 103726.
- Noering, F., Schroeder, Y., Jonas, K., Klawonn, F., 2021. Pattern discovery in time series using autoencoder in comparison to nonlearning approaches. *Integr. Comput.-Aided Eng.* 28, 237–256.
- Noor, S.T., Asad, S.T., Khan, M.M., Gaba, G.S., Al-Amri, J.F., Masud, M., 2021. Predicting the risk of depression based on ECG using RNN. *Comput. Intell. Neurosci.* 2021, <http://dx.doi.org/10.1155/2021/1299870>.
- Pereira, J., Silveira, M., 2019. Unsupervised representation learning and anomaly detection in ECG sequences. *Int. J. Data Min. Bioinform.* 22 (4), 389–407.
- Rajkumar, A., Ganesan, M., Lavanya, R., 2019. Arrhythmia classification on ECG using deep learning. In: *Proceedings of 57th International Conference on Advanced Computing & Communication Systems*. ICACCS, IEEE, pp. 365–369.
- Ren, X., Li, X., Ren, K., Song, J., Xu, Z., Deng, K., Wang, X., 2021. Deep learning-based weather prediction: a survey. *Big Data Res.* 23, 100178.
- Singh, S., Pandey, S.K., Pawar, U., Janghel, R.R., 2018. Classification of ECG arrhythmia using recurrent neural networks. *Procedia Comput. Sci.* 132, 1290–1297. <http://dx.doi.org/10.1016/j.procs.2018.05.045>.
- Sivaraks, H., Ratanamahatana, C.A., 2015. Robust and accurate anomaly detection in ECG artifacts using time series motif discovery. *Comput. Math. Methods Med.* 2015.
- Wang, Y., Yao, H., Zhao, S., 2016. Auto-encoder based dimensionality reduction. *Neurocomputing* 184, 232–242.
- Zhai, X., Zhou, Z., Tina, C., 2020. Semi-supervised learning for ECG classification without patient-specific labeled data. *Expert Syst. Appl.* 158, 113411. <http://dx.doi.org/10.1016/j.eswa.2020.113411>.
- Zhang, C., Li, S., Zhang, H., Chen, Y., 2019. VELC: A new variational autoencoder based model for time series anomaly detection. *CoRR*, [arXiv Preprint myeohosthttp://arxiv.org/abs/1907.01702](https://arxiv.org/abs/1907.01702)[arXiv:1907.01702], <http://arxiv.org/abs/1907.01702>.
- Zhang, Y., Yue, L., 2017. Clustering analysis of ECG data streams. In: *Proceedings of International Conference on Swarm Intelligence*. Lecture Notes in Computer Science (Springer), pp. 304–311.